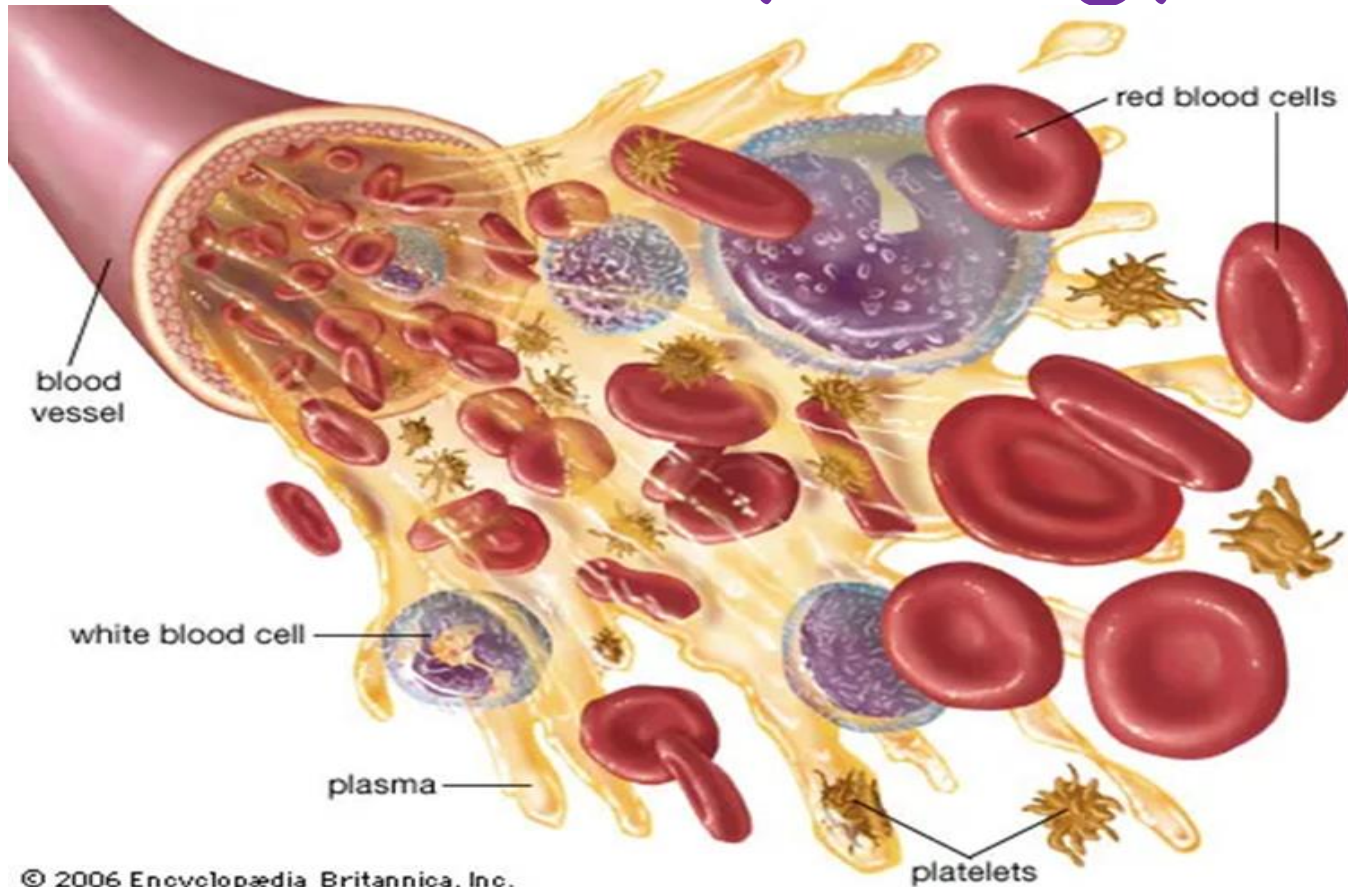


Blood Physiology



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Objectives

At the end of this lesson the students are expected to:

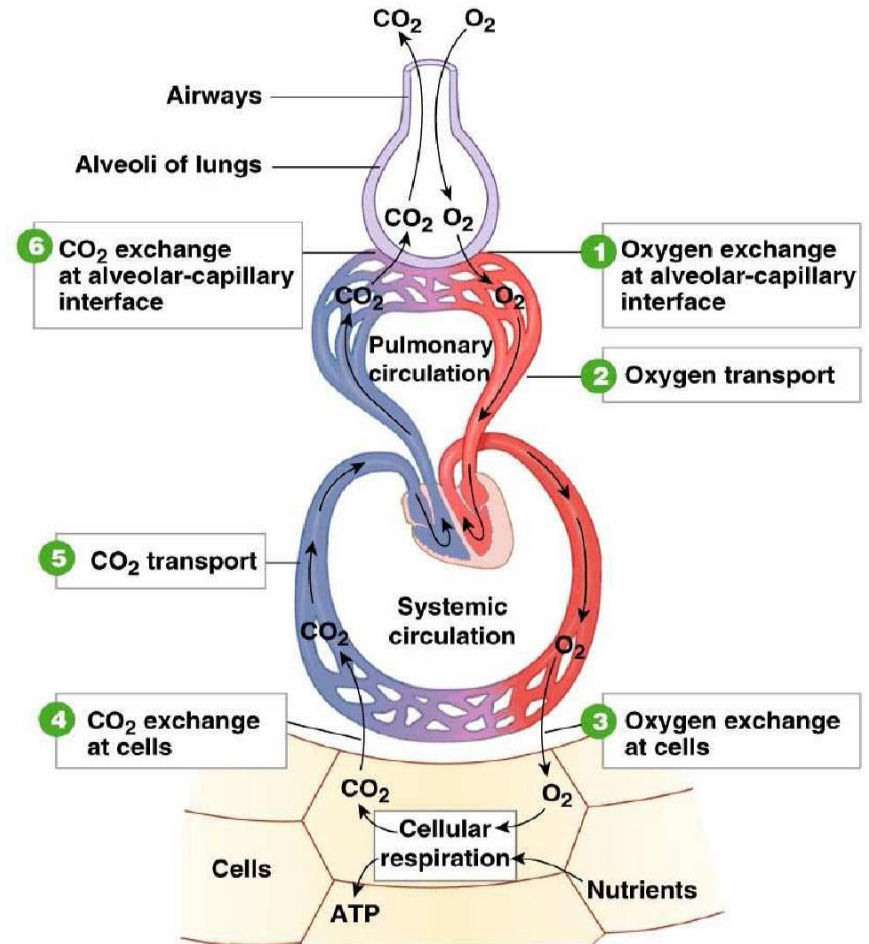
1. Understand the four major functions of blood: transport, hemostasis, homeostasis, and immunity
2. Explain the components of whole blood.
3. Describe the principles and general uses of different laboratory test associated with blood.
4. Discuss about the clinical usefulness of blood typing
5. Clarify the process of hemostasis.

Outlines

- Introduction
- Blood composition
- Functions of blood
- Blood plasma
- Formed elements (cells) in blood
- Human blood types (group)
- Hemostasis

Introduction

- **Blood**, is a connective tissue in fluid form
- Considered as the ‘**fluid of life**’, ‘**fluid of growth**’ and ‘**fluid of health**’.
 - provides nourishment to all body organs & tissues & carries away waste materials.
 - “**the river of life**”, *blood is pumped from heart through network of blood vessels, circulatory system.*



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Introduction ...

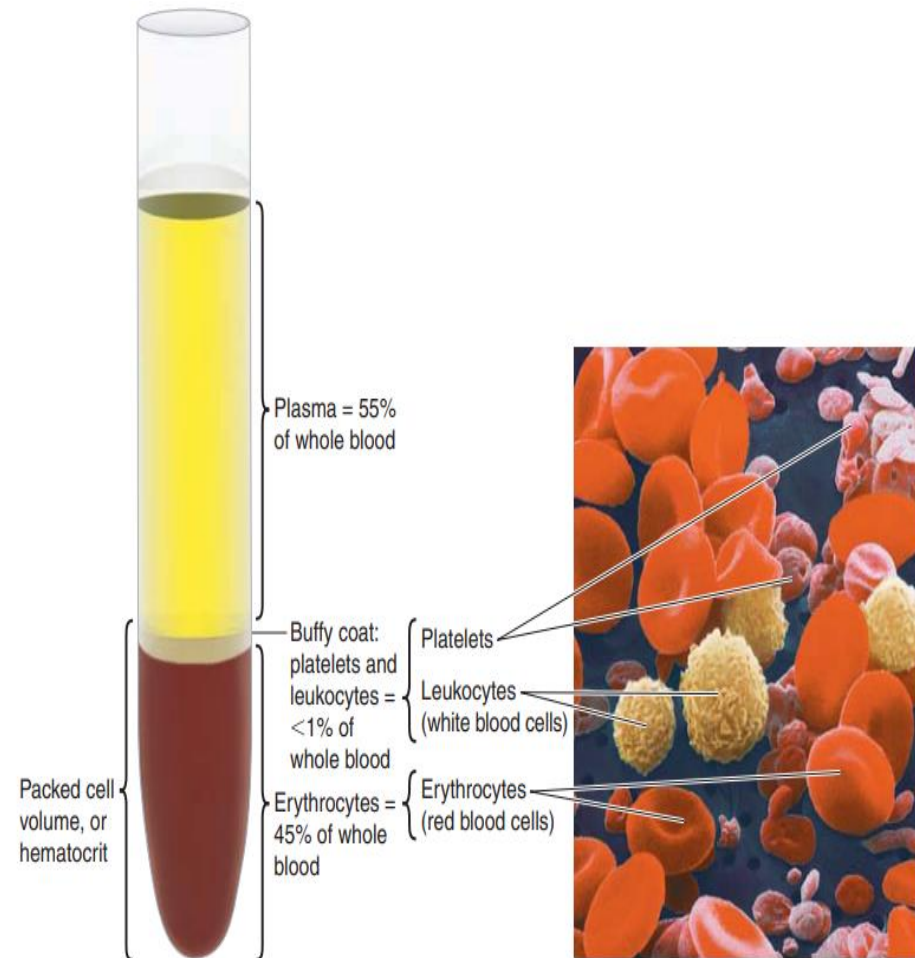
- Adult human has about **5-6 liters** of blood, 7- 8% of total body weight.
- **Infants & children** have comparably **lower volumes of blood**,
 - roughly proportionate to their **smaller size**.
- The **volume of blood** in individual **fluctuates**.
 - Hypovolemia can occur during dehydration, running a marathon, hemorrhage, water loss or sodium depletion...
 - Hypervolemia can occur due to renal failure, excessive sodium intake, or **pregnancy**, when the mother's blood needs to carry **extra oxygen** and **nutrients** to the baby...

Physical Characteristics & Volume

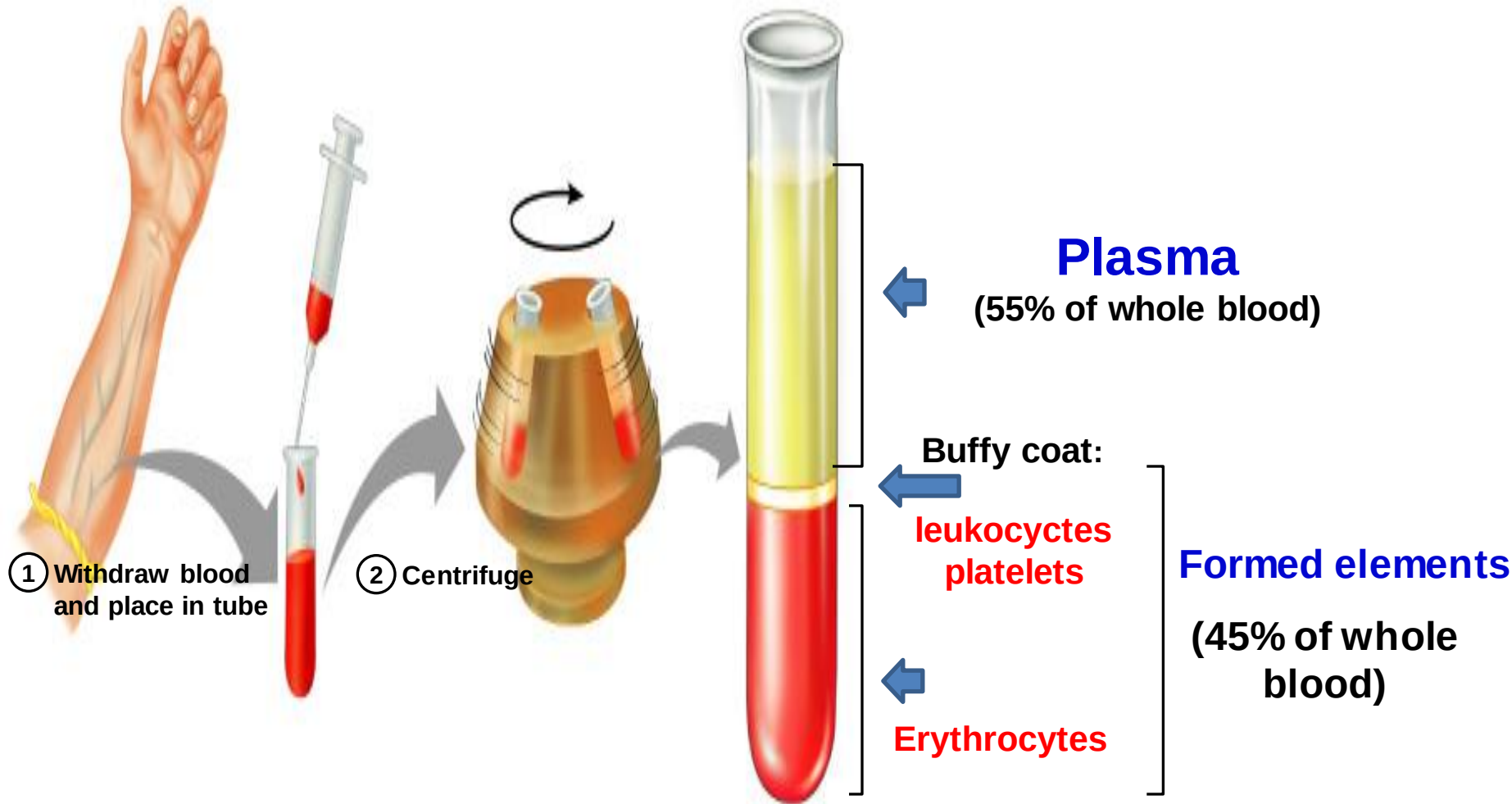
- Blood is **sticky, opaque** fluid with **metallic taste**.
- Color varies from **scarlet** (oxygen-rich) to **dark red** (oxygen-poor).
- The pH of blood is **7.35 – 7.45**, which is slightly alkaline
- Temperature is **37°C (98.6° F)**, which is similar to “normal” body temperature.
- Average volume of blood is **5-6 L** for **males**, and **4-5 L** for **females**.
- *Viscosity*: Blood is five times more viscous than water. It is mainly due to red blood cells and plasma proteins.

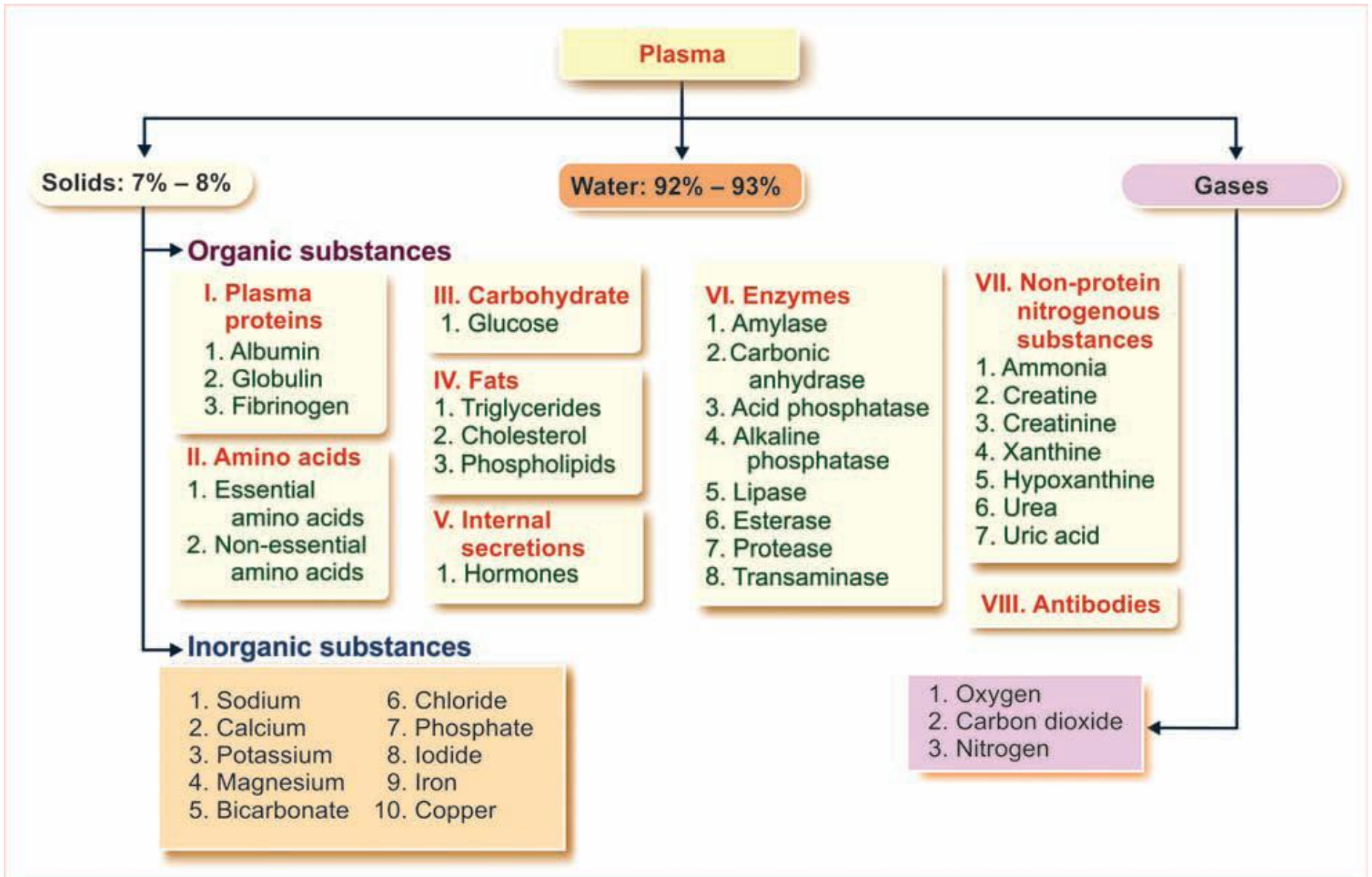
Composition of Whole Blood

- Blood is the body's only fluid tissue composed of:
 - **liquid plasma** (55%) and
 - **formed elements** (45%)
- Formed elements include:
 1. **Erythrocytes** = Red blood cells (RBCs)
 2. **Leukocytes** = White blood cells (WBCs)
 3. **Thrombocytes** = Platelets

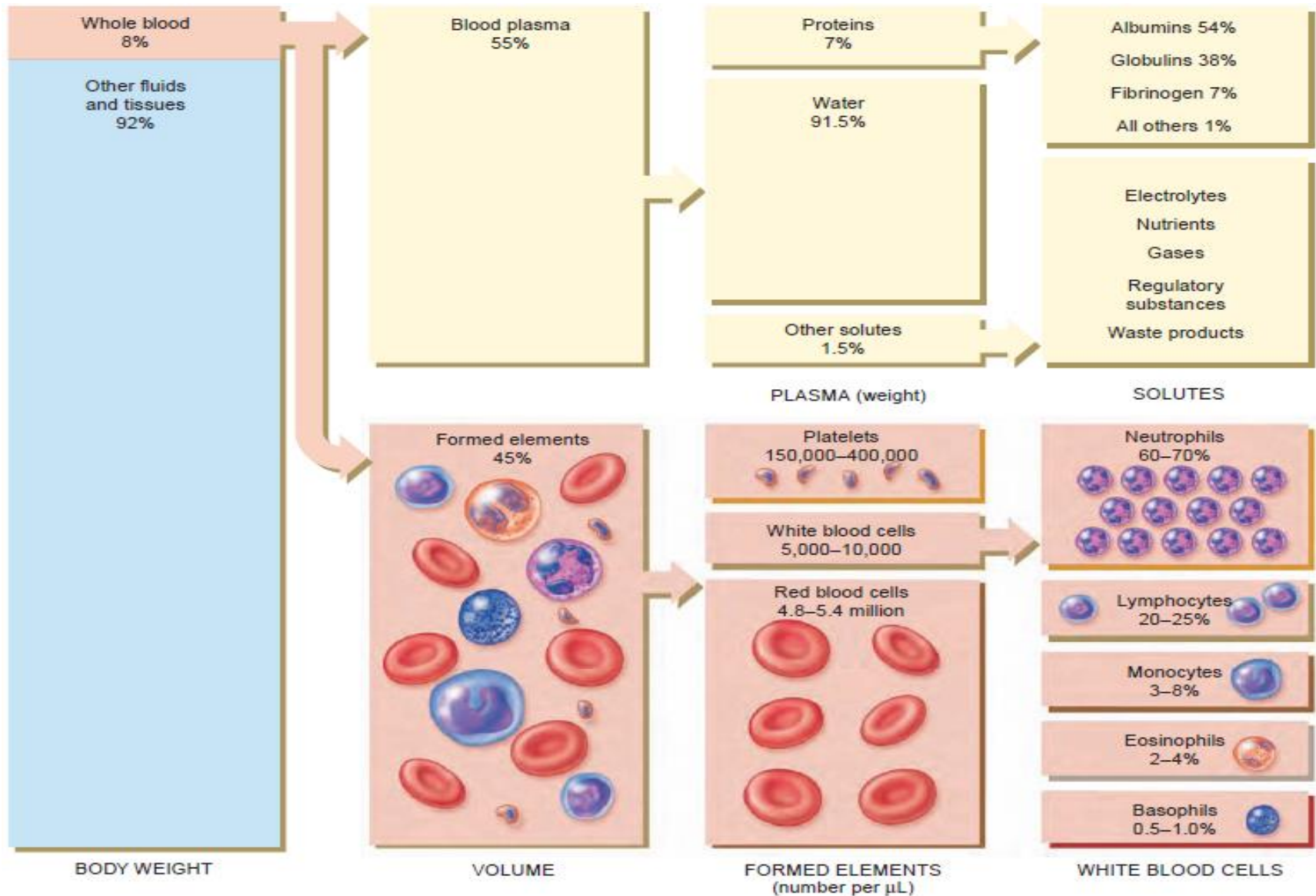


Composition of Blood...





Composition of Blood...



Function of Blood

1. **Distribution**/transportation of substance.
2. **Homeostasis(Regulation)** of pH, temperature & fluid volume.
3. **Protection** of blood loss & infection.

Functions of Blood

1. Transports oxygen, carbon dioxide, nutrients, hormones, heat, and wastes.
2. Regulates pH, body temperature, and water content of cells.
3. Protects against blood loss through clotting, and against disease through phagocytic white blood cells and proteins such as antibodies, interferons, and complement.

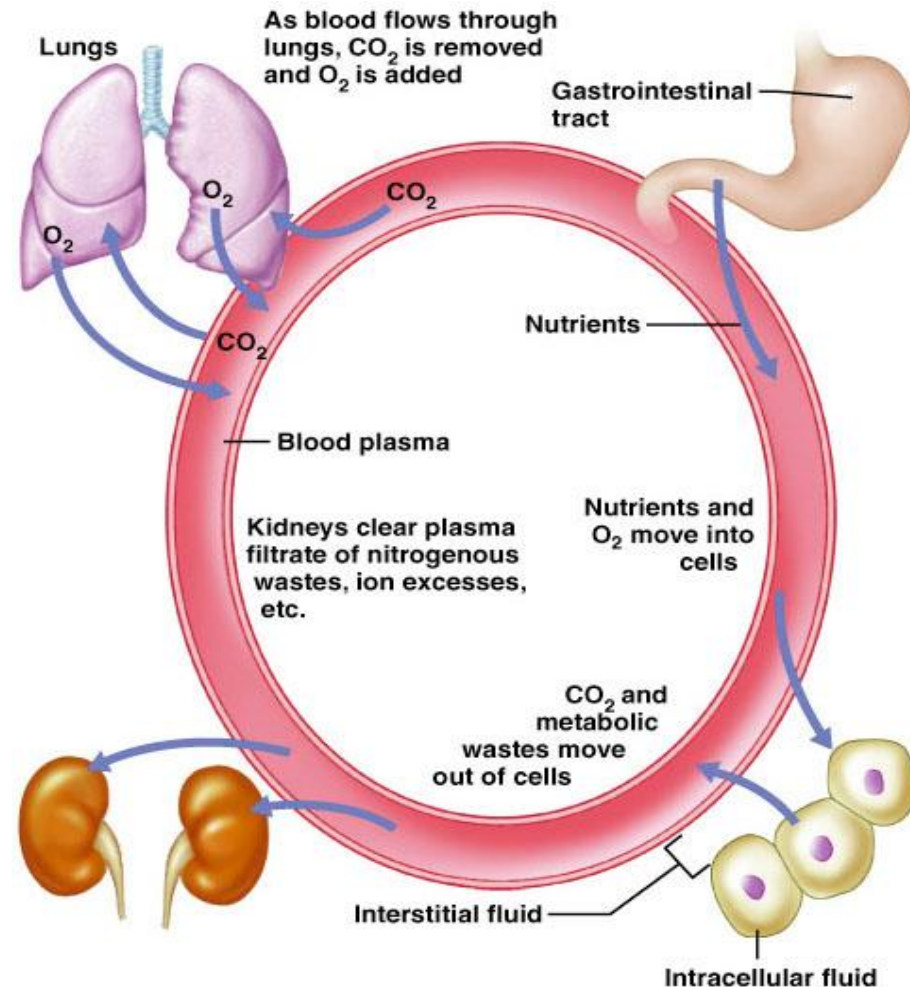
Function of Blood...

- Distribution/transportation.
 - ✓ Blood is the primary means of long-distance transport in the body.
 - ✓ It carries an abundance of important substances from one area of the body to another,
 - including antibodies, acids and bases, ions, vitamins, cofactors, hormones, nutrients, lipids, pigments, metabolites, and minerals.

Distribution...

Blood transports:

1. **Oxygen** from the lungs & **nutrients** from the digestive tract.
2. **Metabolic wastes** from cells to the lungs & to kidneys for elimination.
3. **Hormones** from endocrine glands to target organs.



Function of Blood...

- Hemostasis(arrest of bleeding)

- Blood prevents blood loss by:

- ✓ Activating plasma proteins & platelets.

- ✓ Initiating clot formation when a vessel is broken.

- ✓ The failure to stop bleeding after injury is called hemorrhage and can quickly lead to bleeding to death.

Function of Blood...

- Immunity
 - Blood leukocytes are involved in the body's battle against infection by microorganism
 - Blood prevents infection by:
 1. actually destroying invading bacteria or viruses by **phagocytosis** and
 2. **forming antibodies and sensitized lymphocytes** that may destroy or inactivate the invader

In most cases, the blood's defense system is efficient enough to eliminate the pathogens or to prevent their spreading before they can cause substantial bodily harm.

Function of Blood...

- **Homeostasis.**

- ✓ is a **steady state** that provides an **optimal internal environment** for cell function.
- ✓ The blood system plays a crucial role in preserving homeostasis:
 - By maintaining body temperature, pH, ion conc., osmolality, nutrient supply, and vascular integrity.
- ✓ **Homeostasis** is the result of normal functioning of the **blood's transport, immune, and hemostatic systems.**

Genesis of Blood Cells (**Hematopoiesis**)

- Before birth:
 - first occurs in **yolk sac** of embryo &
 - During the middle trimester of gestation, occurs in **liver, spleen, thymus & lymph nodes** of fetus.
- Last 3 months before birth, **red bone marrow**
 - becomes 1^{ry} site & continues as the source of blood cells after birth & throughout life.
- Red bone marrow is highly vascularized spongy bone tissue present chiefly in bones of;
 - axial skeleton, pectoral & pelvic girdles & proximal epiphyses of humerus & femur.

Hematopoiesis...

- Red bone marrow of essentially **all bone produces** blood cells until a person is **5 years** old.
- After the age **20yrs**, red marrow of long bones becomes quite fatty (yellow marrow) & produces no more blood cells,
 - exceptions: proximal portions of humeri, femur & tibiae.
- After 20yrs blood cells are produced in marrow of **membranous bones** of;
 - cranium, vertebrae, sternum, ribs & pelvic bones.

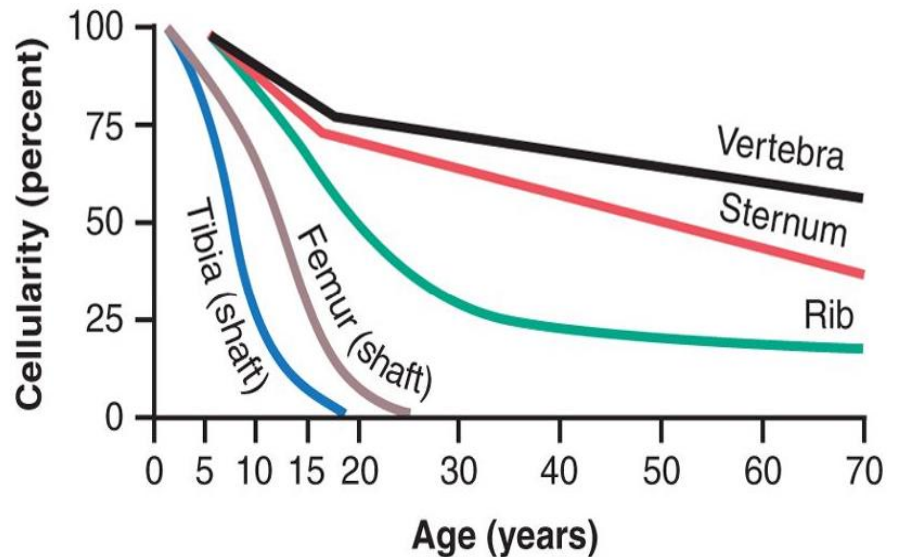


Fig. 3 . Changes in red bone marrow cellularity in various bones with age.

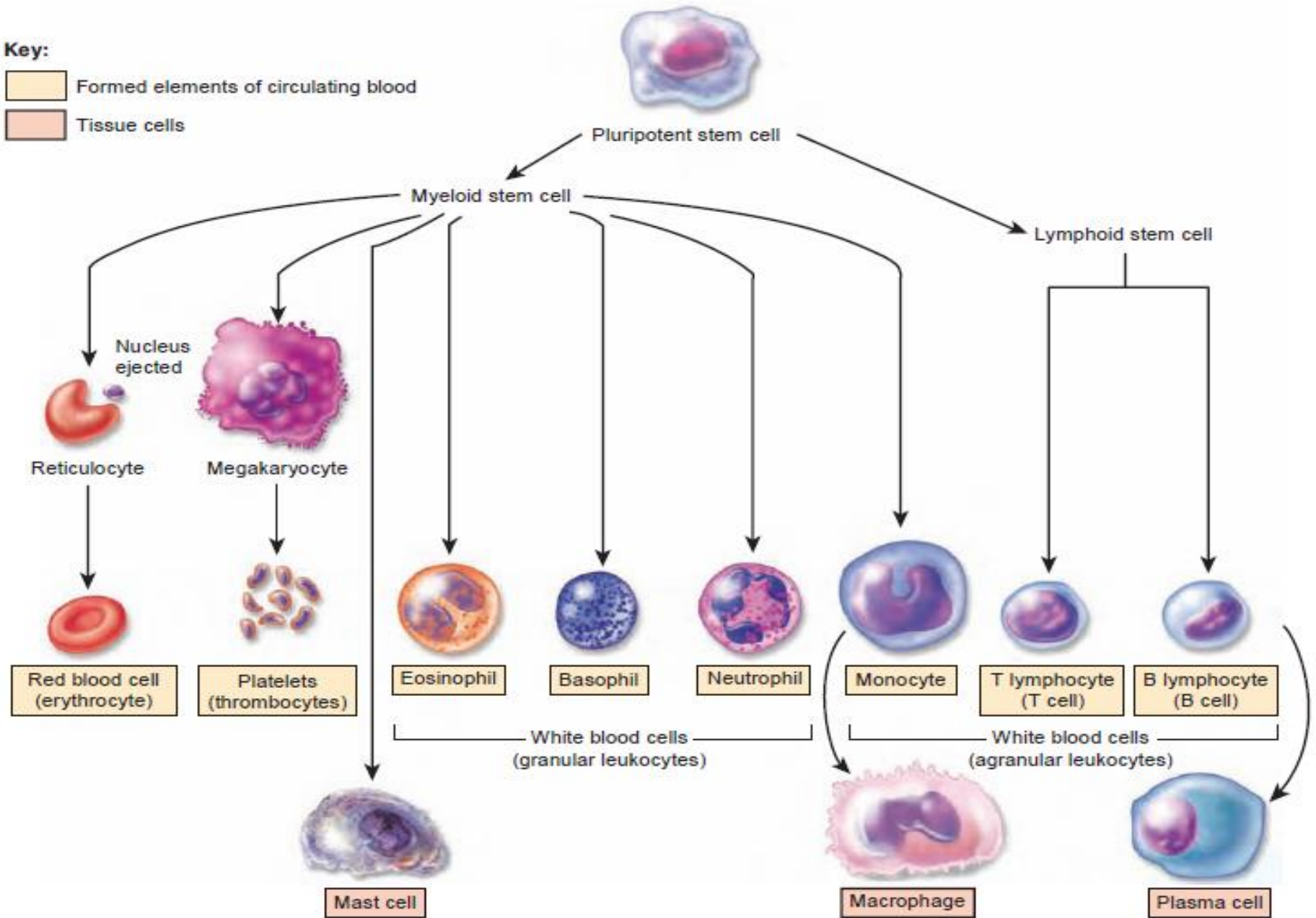
Pluripotent hematopoietic stem cell(PHSC)

- Single type of cell from which all the cells of the circulating blood are eventually derived.
- Capable to reproduce and develop into:
 - myeloid stem cells & lymphoid stem cells.
- **Myeloid stem cells** begin their development in red bone marrow & differentiate into;
 - RBCs, platelets, granulocytes(eosinophils, basophils, neutrophils) & monocytes.
- **Lymphoid stem cells** begin their development in red bone marrow, complete in lymphatic tissues & differentiate into;
 - T and B lymphocytes.

Key:

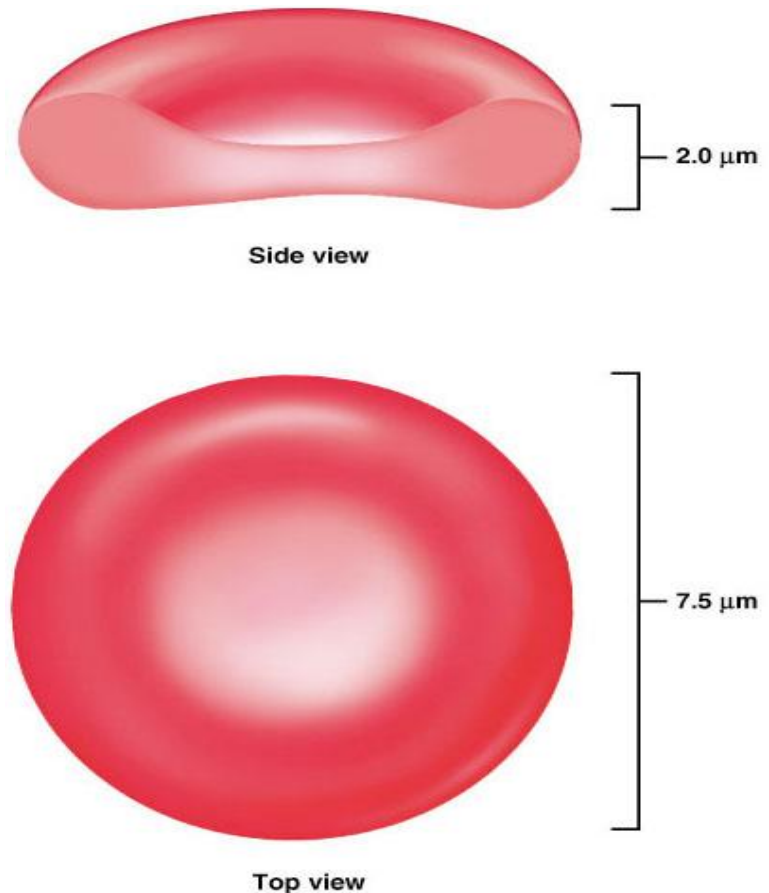
 Formed elements of circulating blood

 Tissue cells



Erythrocytes (RBCs)

- Biconcave discs, anucleate, essentially **no organelles**.
- Filled with **hemoglobin** (Hb),
 - *protein that functions in **gas transport**.*
- Plasma membrane is comprised of *flexible proteins*
 - *allow them to change shape as necessary.*

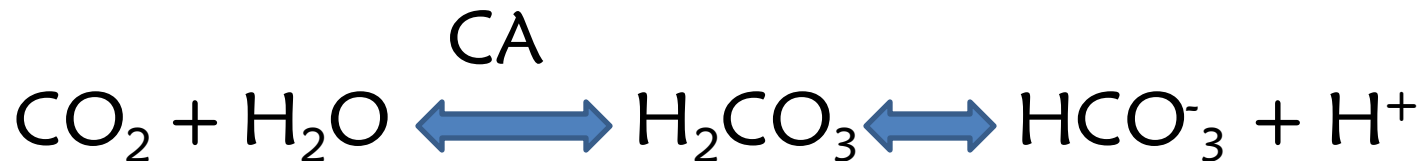


Erythrocytes (RBCs)...

- Are example of how **structure** drives **function**, structural characteristics contribute to its gas transport function:
 - ✓ **Biconcave shape** that has **huge surface area** relative to volume.
 - ✓ **Discounting water content**, about 95% of the erythrocytes are composed of **hemoglobin**.
 - ✓ **ATP** is generated **anaerobically**, erythrocytes do not consume the oxygen they transport.

Erythrocyte Function

1. Transports respiratory gases
 - O_2 & CO_2
2. Hemoglobin serve as a buffer
 - pH balance
3. Carbonic anhydrase (CA)
 - enzyme located in RBC membrane



- CA increases the rate of this reaction 5000 fold

Erythropoiesis (Production of Erythrocytes)

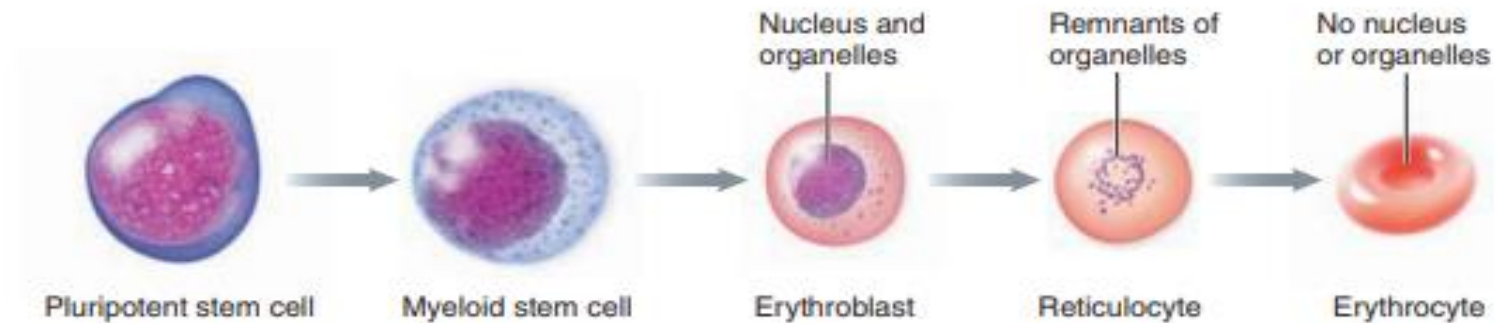
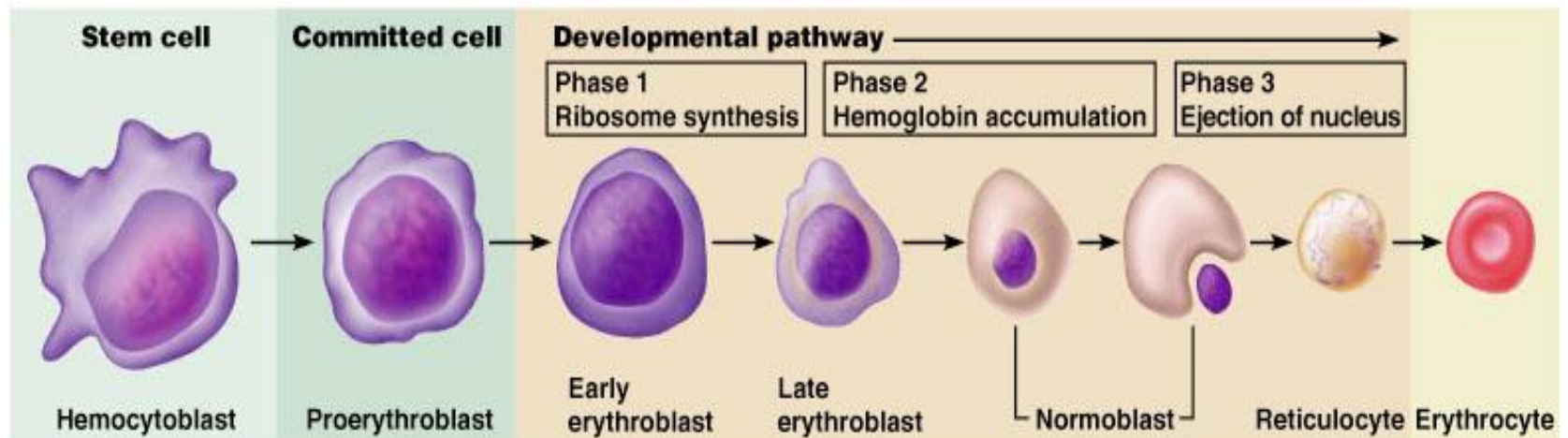


Fig. Major steps in erythrocyte production (erythropoiesis)



Regulation & Requirements for Erythropoiesis

- The total volume of circulating erythrocytes remains remarkably constant because of **reflexes that regulate bone marrow's** production of these cells.
 - Too many RBCs causes  blood viscosity.
 - Too few RBCs leads to tissue hypoxia.
- Erythropoiesis is **hormonally** controlled & depends on adequate supplies of: **iron, vita B₁₂ & folic acid**
 - animal products, green vegetables, fruits, liver, meat.

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TABLE 14.1 | Dietary Factors Affecting Red Blood Cell Production

Substance	Source	Function
Vitamin B ₁₂ (requires intrinsic factor for absorption via small intestine)	Absorbed from small intestine	DNA synthesis
Iron	Absorbed from small intestine; conserved during red blood cell destruction and made available for reuse	Hemoglobin synthesis
Folic acid	Absorbed from small intestine	DNA synthesis

- **Tissue Oxygenation** is the most essential regulator of Red Blood Cell production.
 - ✓ Conditions that decrease oxygen transported to the tissues ordinarily **increase the rate of RBC production.**

Thus,

- ✓ Anemia as a result of hemorrhage or any other condition,
- ✓ **Various diseases** that decrease tissue blood flow, like **prolonged cardiac failure and lung diseases**
- ✓ **very high altitudes**, where the quantity of oxygen in the air is greatly decreased.

all these conditions cause low oxygen supply to tissue, stimulating the kidney for the secretion of **Erythropoietin(EPO)**

Hormonal Control of Erythropoiesis

- Normally, about 90% of all EPO is formed in the kidneys, and the remainder is formed mainly in the liver
- EPO production by the **kidneys** is triggered by **hypoxia** due to:
 1. Decreased RBCs count.
 2. Decreased oxygen availability.
 3. Increased tissue demand for oxygen.
- EPO stimulate the proliferation of **erythrocyte progenitor cells** and their differentiation into mature erythrocytes, **which finally increase**
 - Number of RBC in circulating blood and
 - Oxygen carrying capacity of the blood and restoring O₂ delivery to the tissues.

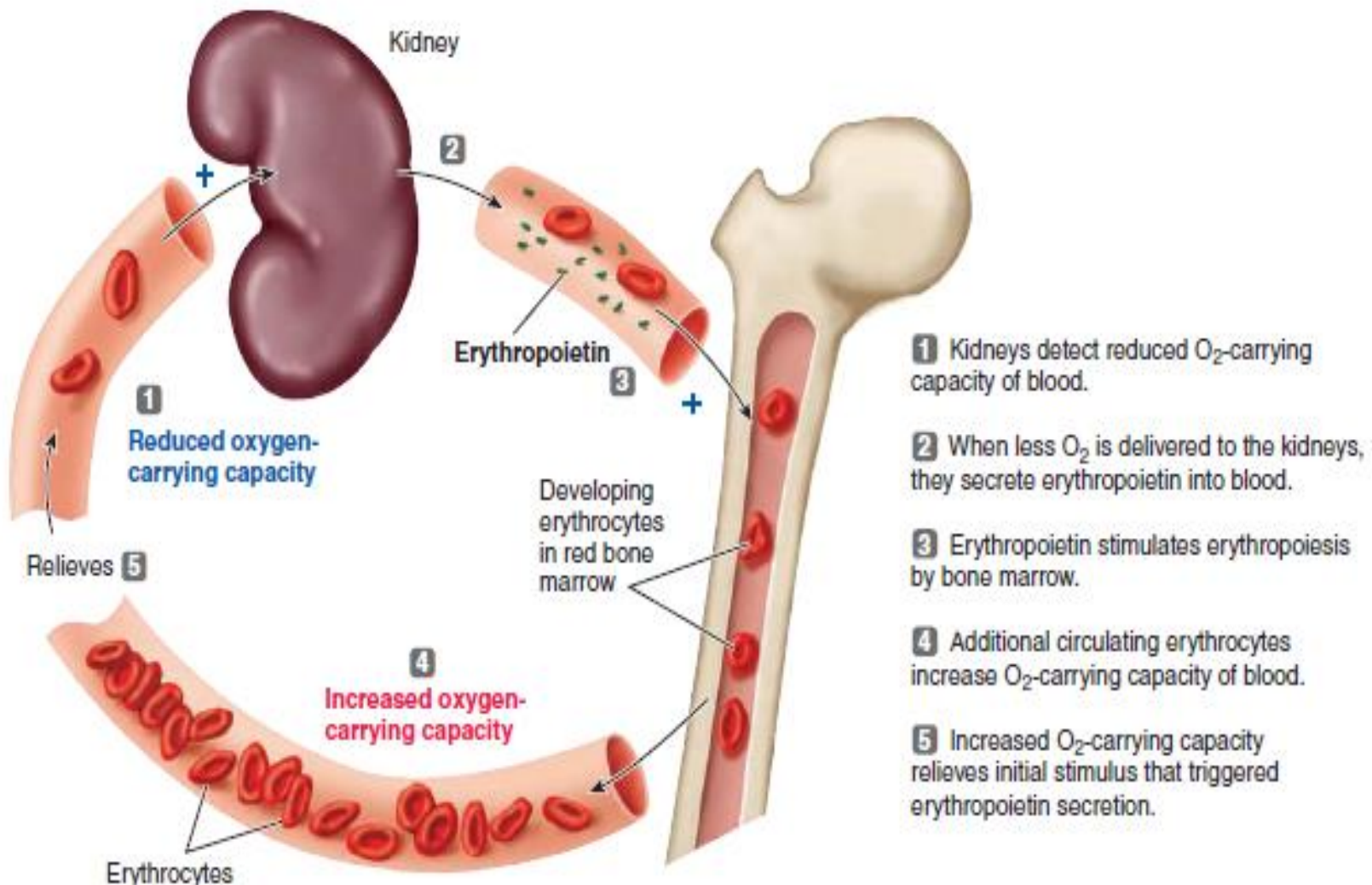
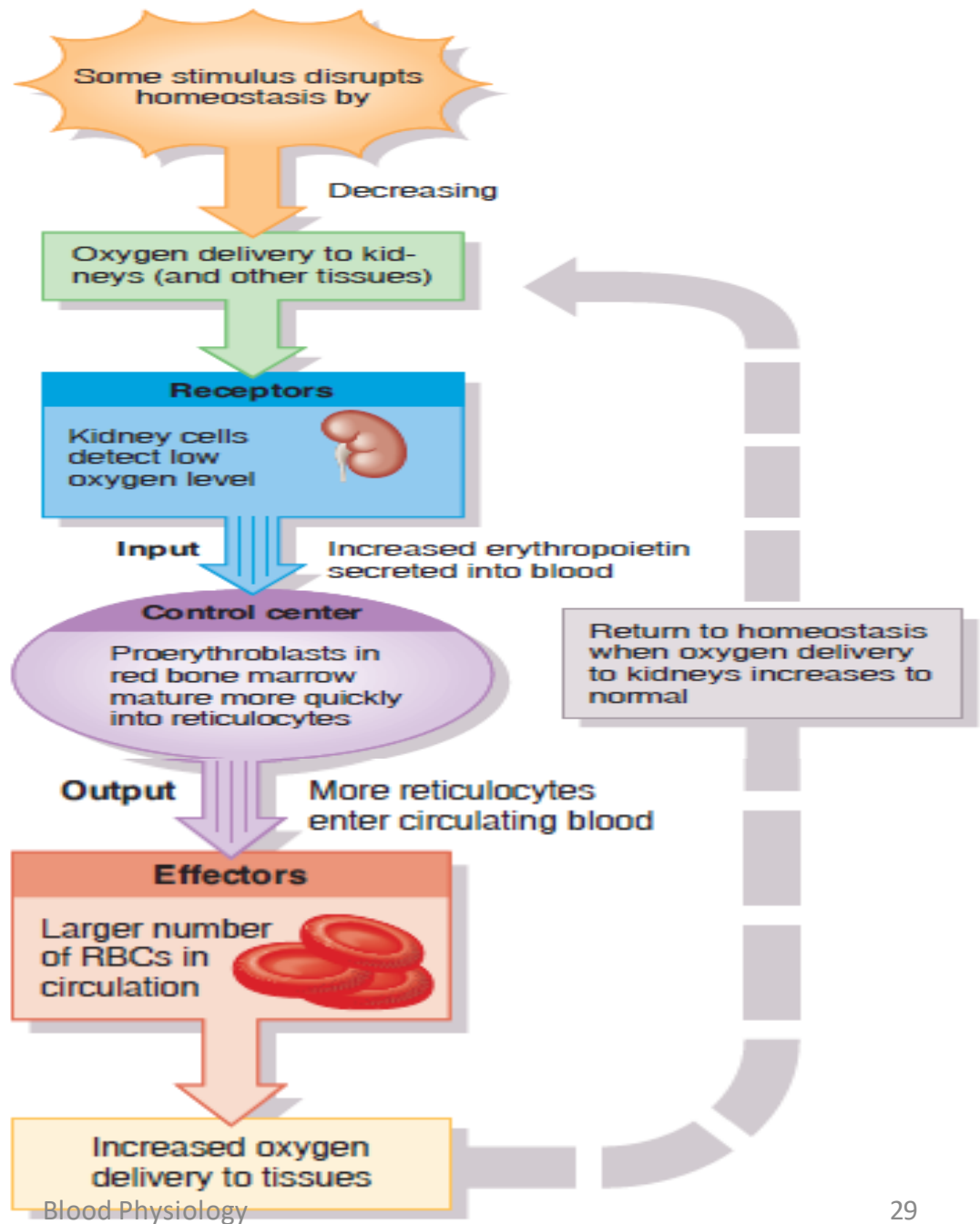
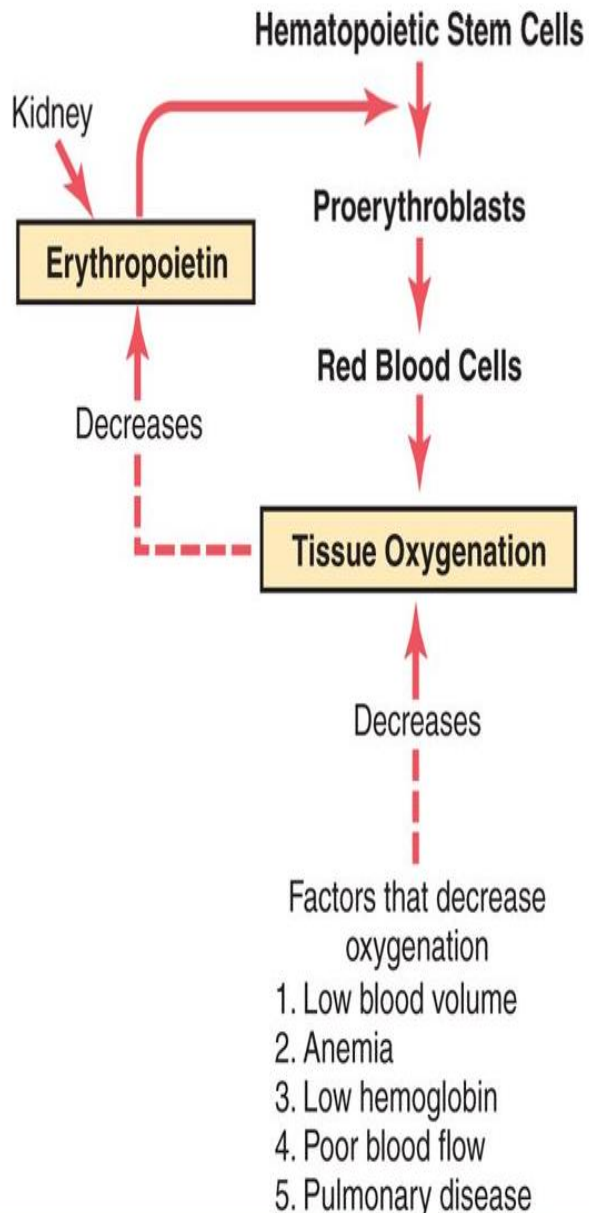
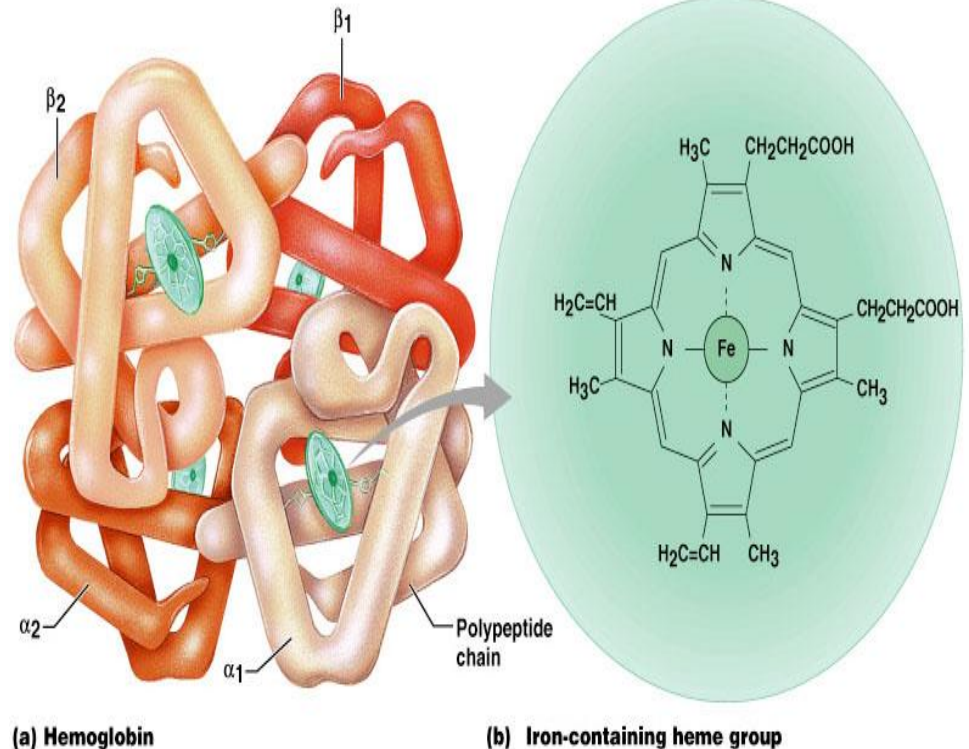


Figure . Decreased oxygen delivery to the kidneys increases erythrocyte production via increased erythropoietin secretion



Hemoglobin

- Reversibly binds with most O_2 in the blood.
- Composed of 4 protein **globin** chains of each bound to **heme** group,
 - each heme bears an atom of **iron**, which can bind to one O_2 molecule.
 - each Hb molecule can transport 4 molecules of O_2 .



Hemoglobin structure

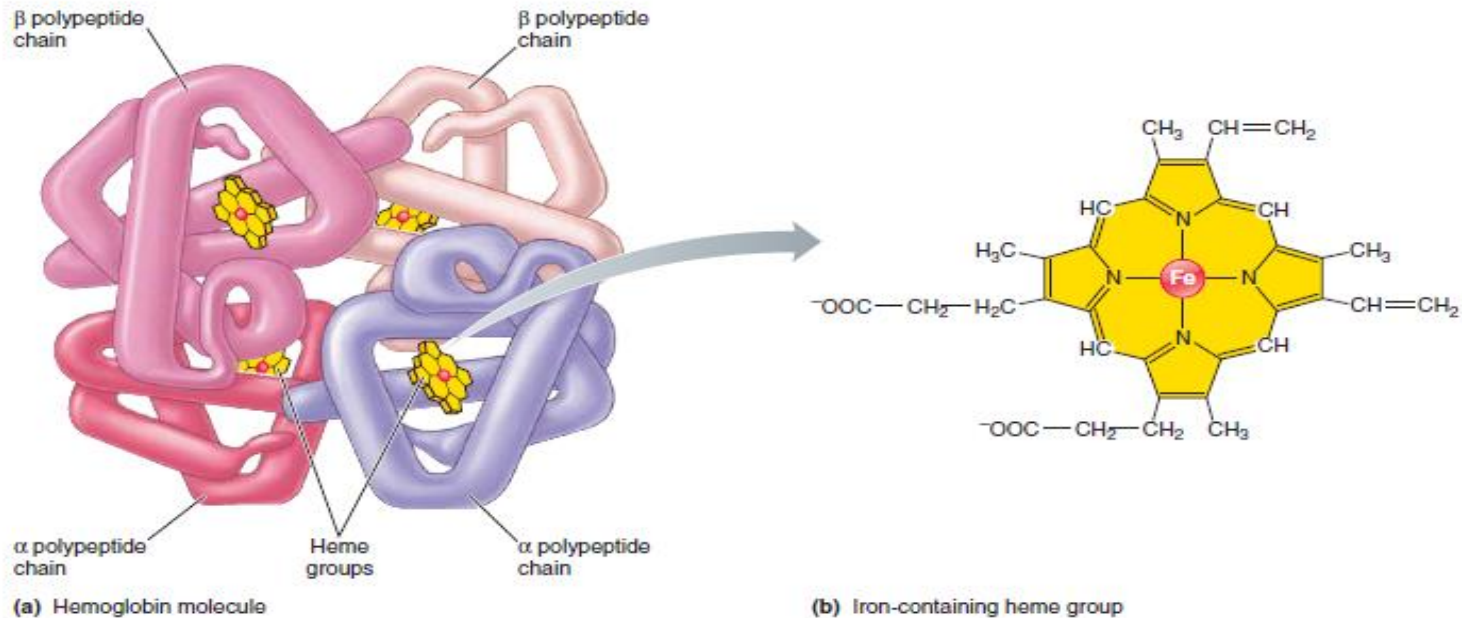


Fig. A hemoglobin molecule consists of four highly folded polypeptide chains and four iron-containing heme groups.

In the lungs, hemoglobin picks up oxygen, which binds to the iron ions, forming **oxyhemoglobin**.

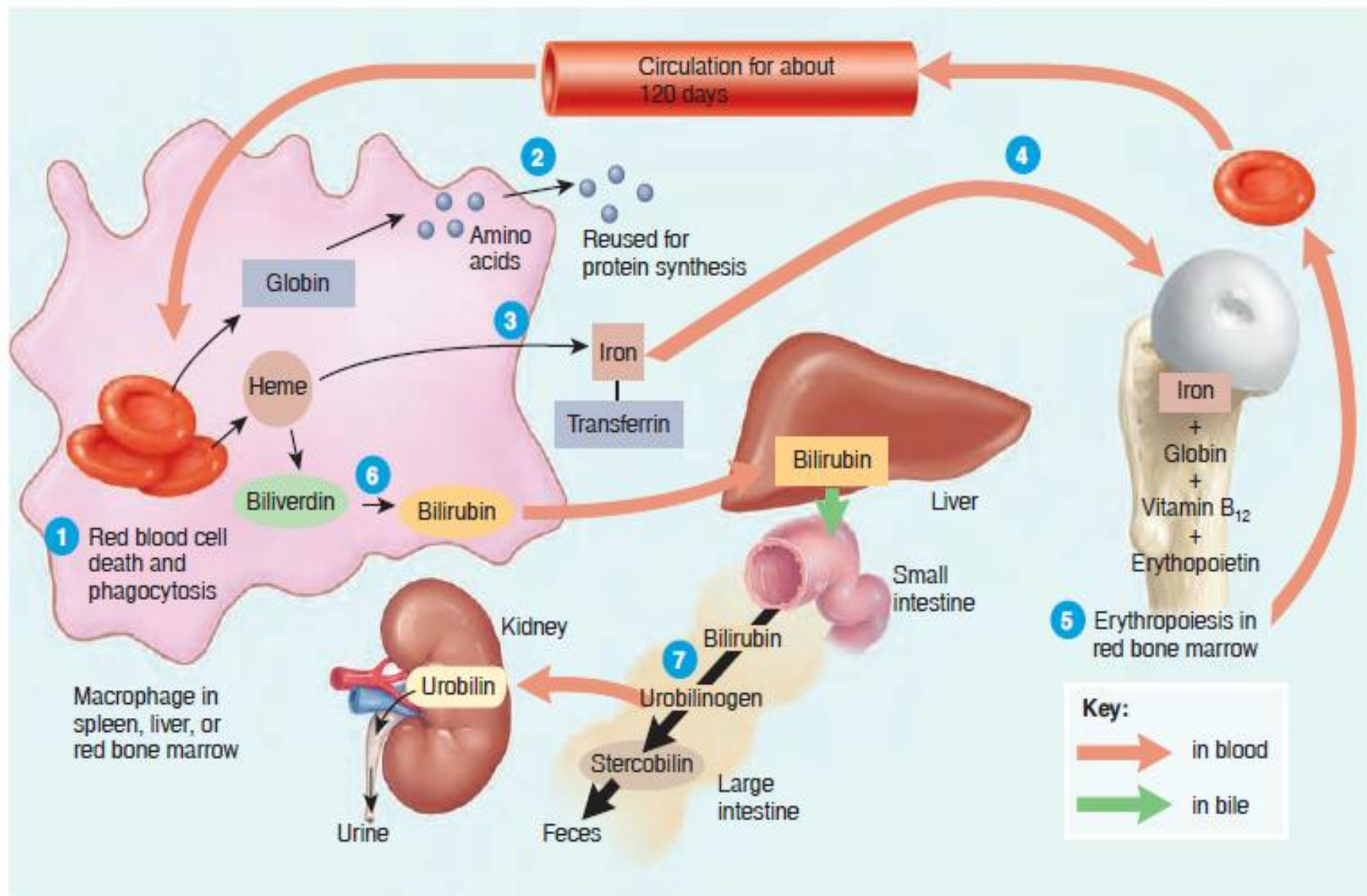
The bright red, oxygenated hemoglobin travels to the body tissues, In the tissue, where it releases some of the oxygen molecules, becoming darker red **deoxyhemoglobin**, sometimes referred to as reduced hemoglobin.

Fate & Destruction of Erythrocytes

- The **life span** of an erythrocyte is 100–120 days.
- Old erythrocytes become rigid and fragile, and their hemoglobin begins to degenerate.
- Dying erythrocytes are **engulfed** by **macrophages**.
- Heme & globin are separated & the iron is salvaged for reuse:
 - **Heme** : broken down into **iron** and **bilirubin**
 - **Globin** : a reusable protein



The rate of RBC formation by red bone marrow equals the rate of RBC destruction by macrophages.



Major Events in Red Blood Cell Destruction

1. Squeezing through the capillaries of active tissues damages red blood cells.
2. Macrophages in the spleen and liver phagocytize damaged red blood cells.
3. Hemoglobin from the red blood cells is decomposed into heme and globin.
4. Heme is decomposed into iron and biliverdin.
5. Iron is made available for reuse in the synthesis of new hemoglobin or is stored in the liver as ferritin.
6. Some biliverdin is converted into bilirubin.
7. Biliverdin and bilirubin are excreted in bile as bile pigments.
8. The globin is broken down into amino acids metabolized by macrophages or released into the blood.

Clinically important lab tests cont'd...

Hemoglobin concentration (Hb)

- amount of Hb in a **given volume of blood** (g/dl), important to diagnose anemia
- vary according to *age, gender & altitude*.
- ❑ **Normal hemoglobin reference range** (guideline figures)
 - **Children** ----» at birth = 13.5 - 19.5 g/dl
 - 2 - 5 years = 11 - 14 g/dl
 - 6 - 12 years = 11.5 - 15.5 g/dl
 - **Adult** -----» men = 13 - 18 g/dl
 - women (non pregnant) = 12 - 15 gm/dl
 - pregnant women = 11 - 13.8 gm/dl
- ❑ Hb level less than the lower cut off point is **anemia**.

Clinically important lab tests cont'd...

Packed cell volume (PCV)/ Hematocrit (Hct)

- Percentage of RBC in given volume of whole blood.
- Hct can be determined by centrifugation of anticoagulated blood within small capillary tube.
- ✓ clinically significant to diagnosis anemia
- On average it is about **45%**
 - in adult male: 39%- 54%
 - in adult women: 36%- 45%
 - pregnant women: 33% - 42%
- Approximately it is the hemoglobin value times 3
- Hct below the lower cut off point indicates **anemia**.

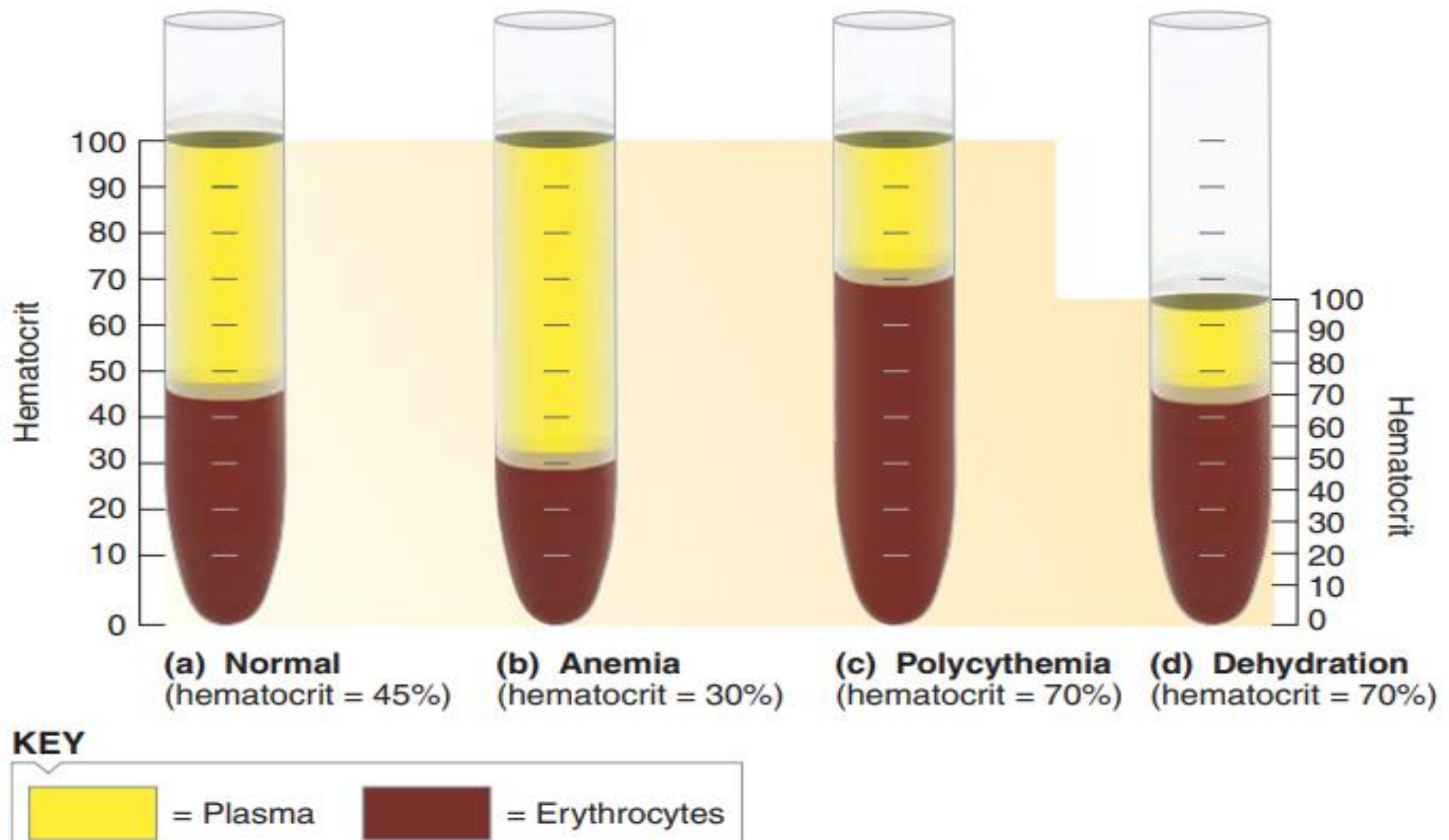


Fig. (a) Normal hematocrit. (b) The hematocrit is lower than normal in anemia, because of too few circulating erythrocytes; and (c) above normal in polycythemia, because of excess circulating erythrocytes. (d) The hematocrit can also be elevated in dehydration when the normal number of circulating erythrocytes is concentrated in a reduced plasma volume

Clinically important lab tests cont'd...

- Complete blood count(CBC) is a more definitive test for blood disorders
- complete blood count (full blood count), is part of a routine medical examination.
- The CBC includes the following tests,
 - ✓ red blood cell (RBC) count,
 - ✓ Hgb, and Hct,
 - ✓ RBC indices;
 - ✓ total white blood cell (WBC) count
 - ✓ differential WBC count;
 - ✓ platelet count and mean platelet volume
 - ❖ Usually done using an automated CBC counting Machine.

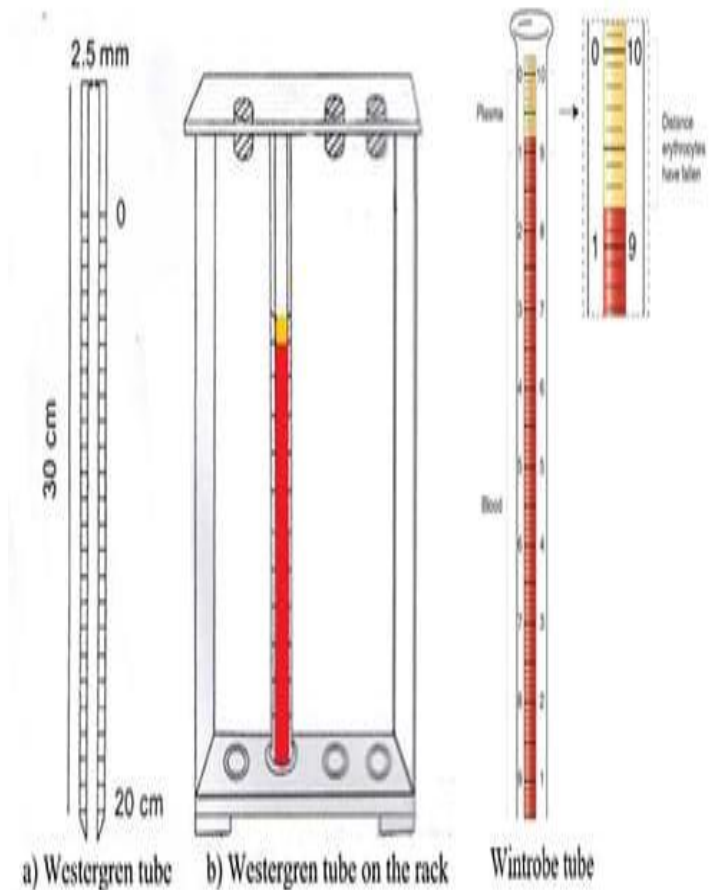
Clinically important lab tests cont'd

Erythrocyte sedimentation rate (ESR)

- ESR is the rate at which uncoagulated blood (RBC) settles down when allowed to stand in a tube.
- Erythrocytes have a slightly higher density than the suspending plasma, so that they slowly settle out of whole blood
- Caused by rouleaux formation- pilling of RBC like coins together
- This test is performed by placing anticoagulated blood in an upright tube (Westgren's or wintrobe's tube),
 - at the end of some specific time(usually 1hr), rate of RBC sedimentation is measured
 - High ESR indicate the present of inflammation.

Erythrocyte sedimentation rate (ESR)...

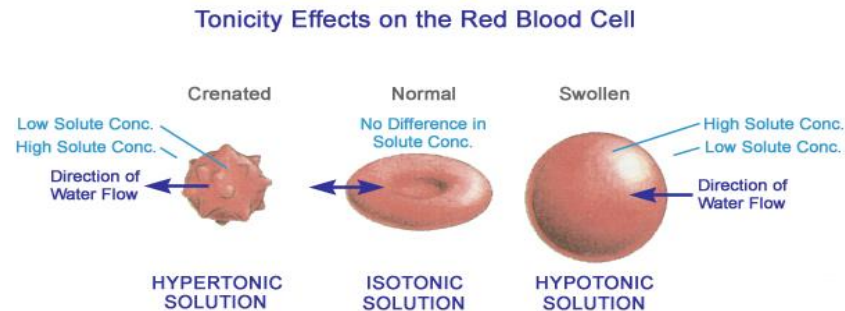
- Reference range
 - adult females 0-20 mm/hr
 - adult males 0-15 mm/hr
- Increased ESR may be due to:
 - pregnancy, inflammation, cancer.
- Decreased ESR may be due to :
 - polycythemia, sickle cell anemia, hereditary spherocytosis, congestive heart failure.



Clinically important lab tests cont'd...

Osmotic fragility test(OFT)- a test to measure red blood cells (RBC) resistance to hemolysis when exposed to hypotonic solutions.

- Cells like RBC shrink or swell in solutions of osmotic pressure greater or less than that of normal plasma.
- ✓ OFT measure the ability of erythrocytes to take up fluid without lysing.
- A 0.9% NaCl solution is isotonic with plasma.



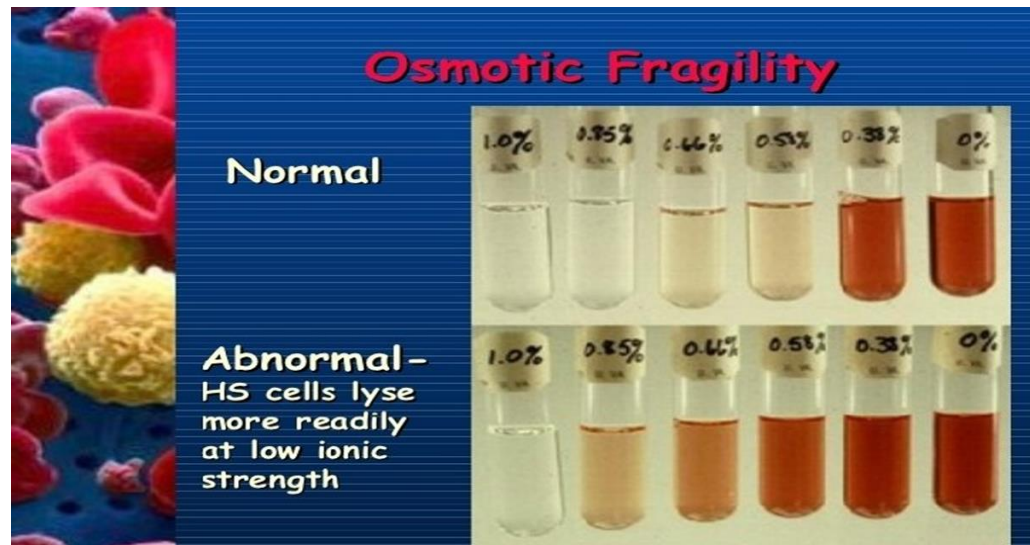
Osmotic fragility test...

- To maintain viable cells in vivo/vitro the cells must be suspended in a solution with **an osmotic conc. relatively equal to their interior osmotic conc.**
- RBC gets - swell then **hemolyzed** in hypotonic (<0.9% NaCl) & **shrink** (crenate) in hypertonic(>0.9% NaCl) solution.
 - the sooner hemolysis occurs, the greater is osmotic fragility of RBC

Cont'd

When osmotic fragility **test is normal**,

- Hemolysis of normal RBC begins at saline conc. of 0.50 – 0.55%
- 50% RBC lyse(MCF) occur at saline conc. of 0.40%-0.45%
- Complete hemolysis occur at saline conc. of $< 0.35\%$



Clinical significance of OFT

- Useful in diagnosis of different types of hereditary hemolytic anemia.
- Increased osmotic fragility
 - hereditary spherocytosis, acquired spherocytosis
- Decreased osmotic fragility
 - Thalassemia, sickle cell anemia, iron deficiency anemia

Human Blood Groups/types

- Grouping depends on: *presence/absence of antigens/agglutinogens on surface of RBCs.*
- Agglutinogens- **protein** present on the RBC membrane with **antigenic** effect (ability to cause the production of antibody against it or **ability to bind specific antibody molecules**)
- Agglutinins/antibodies – are proteins produced against agglutinogens/antigen
 - Agglutination – **reaction between these two**
- Major blood groups are:
 - **ABO blood group & Rh system**

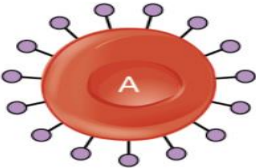
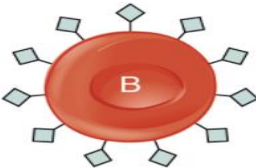
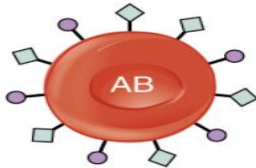
ABO blood grouping/typing

Two antigens(type A and type B) present on the surfaces of RBCs in a large proportion of human beings

Karl Landsteiner's Law

- If an antigen present on the RBC's of an individual, the corresponding antibody must be absent from the plasma.
- If an antigen is absent on the RBC's of an individual, the corresponding antibody must be present in the plasma.

Blood type	Agglutininogen on RBC	Agglutinin in plasma
A	A	Anti-B
B	B	Anti-A
AB	A and B	None
O	None	Anti- A and Anti B

Blood Type				
	A	B	AB	O
Red Blood Cell Type				
Antibodies in Plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in Red blood Cell	 A antigen	 B antigen	 A and B antigens	None
Blood Types Compatible in an Emergency	A, O	B, O	A, B, AB, O (AB ⁺ is the universal recipient)	O (O is the universal donor)

Relative Frequencies of the Different Blood Types.
The prevalence of the different blood types among one group of persons studied was approximately:

O	47%
A	41%
B	9%
AB	3%

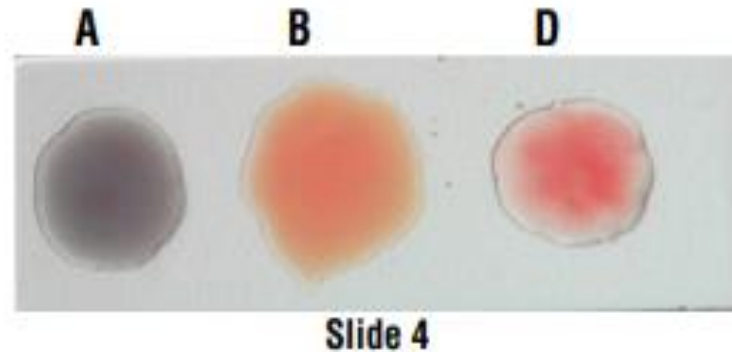
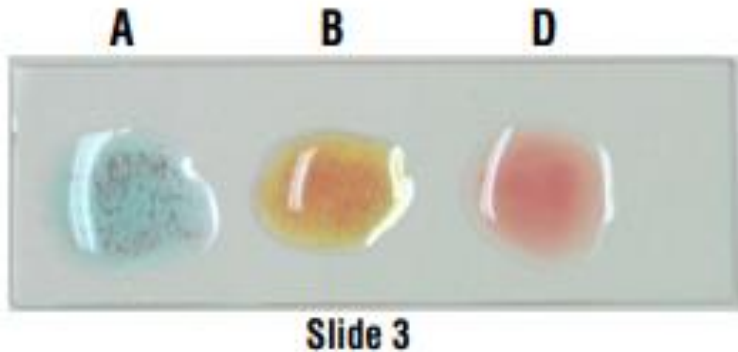
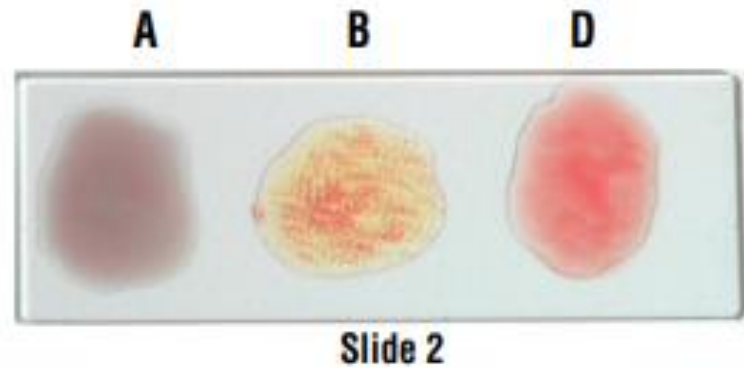
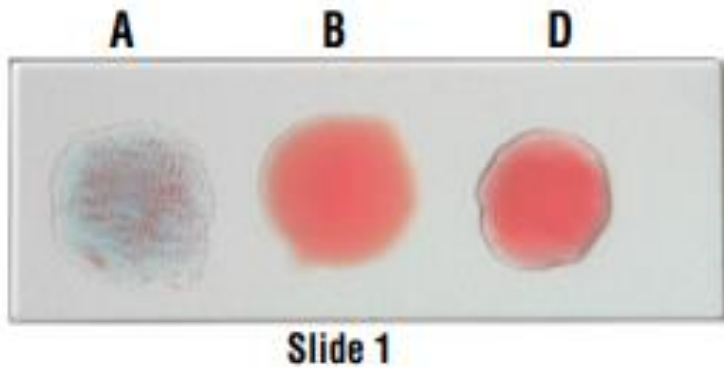
Importance of Knowing Blood groups

- For blood transfusion
- Hemolytic disease of the newborn
- Paternity dispute
- Susceptibility to various diseases (blood group O- peptic ulcer, group A –gastric cancer)

Rh Factor (Rh blood group)

- It was first worked out on **Rhesus monkey** to the name Rh-system.
- individual grouped as **Rh⁺** & **Rh⁻** based on **±** of **agglutinogen D** on the surface of RBCs:
 - **Presence** of agglutinogen D on RBCs = **Rh⁺**
 - **Missing** agglutinogen D on RBCs = **Rh⁻**
- **Agglutinin (anti-D antibodies)** are not normally present in the serum,
 - produced 2^{ry} to exposure of Rh⁻ blood to Rh⁺ blood (antigen-D).

ABO/Rh blood grouping in Lab....

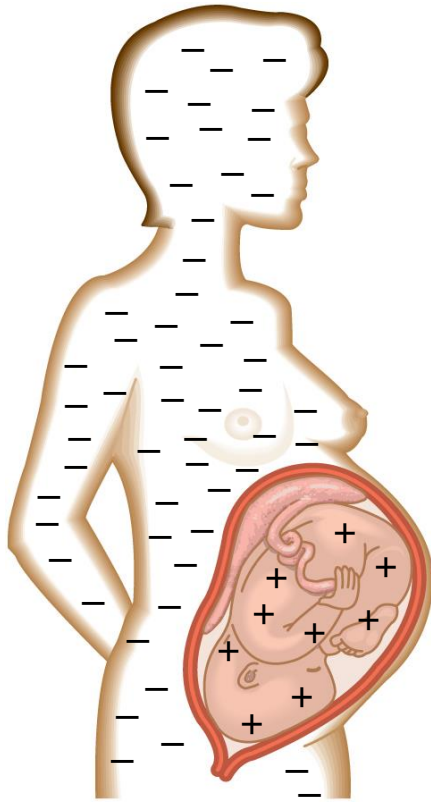


Report as

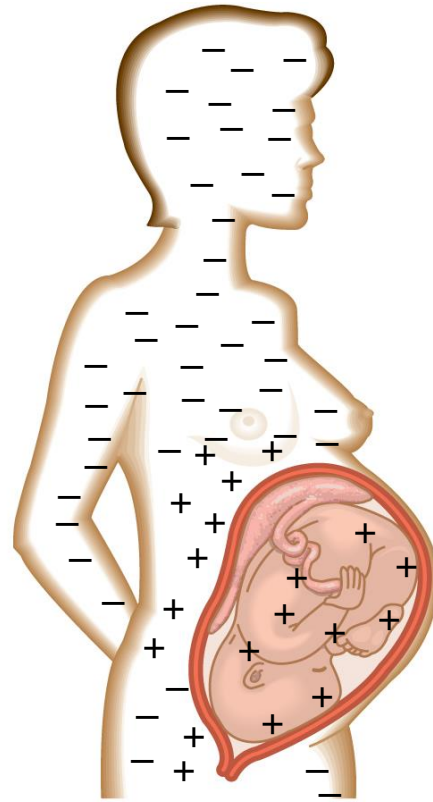
- ✓ Slide 1 A-, Slide 2 B+,
- ✓ Slide 3 AB-, Slide 4 O+

Erythroblastosis Fetalis/HDFN

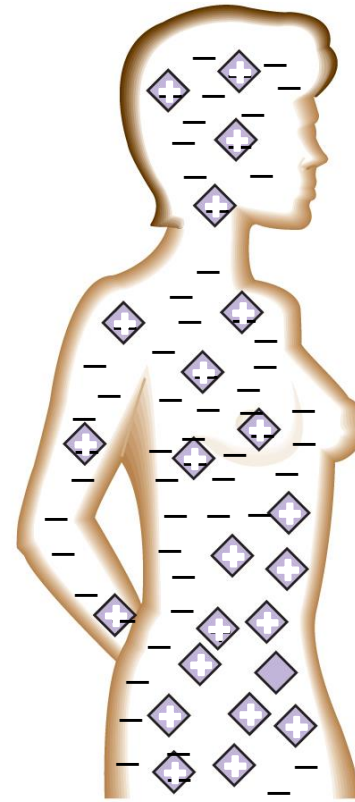
- Incompatibility of Rh blood groups b/n **fetus & mother**
 - ✓ Antibodies produced by Rh⁻ mother destroy RBCs of Rh⁺ fetus.
- occurs when **Rh⁻ mother** marries **Rh⁺ father** & conceives **Rh⁺ fetus**.
- During delivery, there could be leakage of Rh⁺ blood from fetus to circulation of mother.
 - Rh⁺ blood induces production of anti-D antibodies in circulation of mother.
- During **2nd conception** of Rh⁺ fetus, anti-D antibodies cross placenta & attack RBCs of fetus. (Rh incompatibility rxn occur)
- Most population in the world (85%) is Rh⁺
- **Treatment** : Rho Gam injection after 1st birth **blocks mother from forming anti-Rh antibody** [Limits risk for subsequent births].



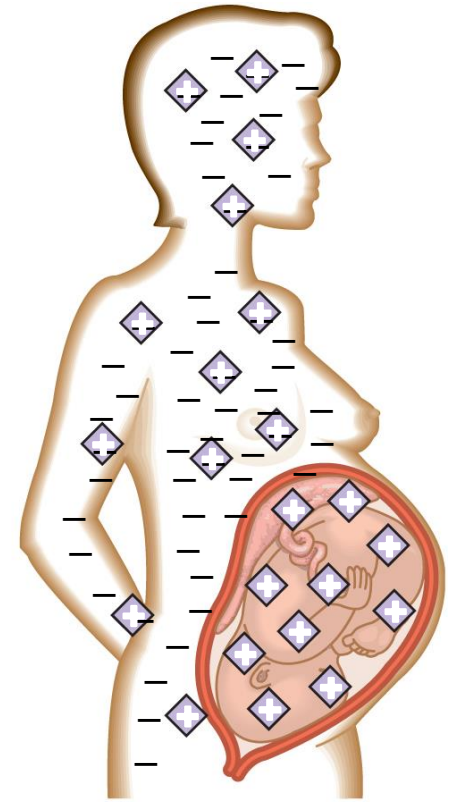
Rh-negative woman with Rh-positive fetus



Cells from **Rh-positive fetus** enter woman's bloodstream



Woman becomes Sensitized, produce antibodies to fight Rh-positive blood cells



In the next Rh-positive pregnancy, maternal antibodies attack fetal RBCs

Blood Transfusions

- **Giving blood** of one person to another when there is a need.
It can be:

1. RBC/ Packed red cells transfusion

- cells with plasma removed, are used to treat **anemia**.

2. Platelet Transfusions

- Platelets stops the body from bleeding.
- Often patients suffering from leukemia, or other types of cancer, have lower platelet counts as a side effect of their chemotherapy treatments..

3. Plasma Transfusions

- Contains important proteins and other substances crucial to one's overall health
- Used for patients with liver failure, severe infections, and serious burns.

Compatibility b/n Donor's Cells & Recipient's Serum

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TABLE 14.13 | Preferred and Permissible Blood Types for Transfusions

Blood Type of Recipient	Preferred Blood Type of Donor	Permissible Blood Type of Donor (In an Extreme Emergency)
A	A	O
B	B	O
AB	AB	A, B, O
O	O	No alternate types

Transfusion Reactions

- Occur when **mismatched blood** is infused.
- **Donor blood cells** are **attacked** by the **recipient's plasma**:
 - Diminished oxygen-carrying capacity.
 - Clumped cells impede blood flow.
 - Ruptured RBCs release **free hemoglobin** into the bloodstream
 - Circulating Hg precipitates in kidneys & causes **renal failure**.

Compatibility b/n Donor's Cells & Recipient's Serum	Donor's Cells	Recipients Serum			
		A	B	AB	O
	A	✓	X	✓	X
	B	X	✓	✓	X
	AB	X	X	✓	X
	O	✓	✓	✓	✓
✓ = No Agglutination , X = Agglutination [clumping of RBCs in the presence of antibody]					

Precautions to be taken while blood donation or transfusion

- Should be **Health**, Screened for – AIDS, viral Hepatitis, malaria, syphilis
- Age- 17- 60Yrs, weigh **over 50kg**
- **Hg and PCV** should be normal
- During transfusion- only compatible blood must be transfused.
 - Both **major & minor cross matching** of recipient & donor blood must be done.
 - ✓ Major - Donor's RBC + Recipient plasma
 - ✓ Minor – Donor's plasma +Recipient RBC
- Rh compatibility must be confirmed(Rh+ blood never be transfused to Rh –ve person)
- Contraindicated in pregnant and lactating mothers

Leucocyte/WBCs

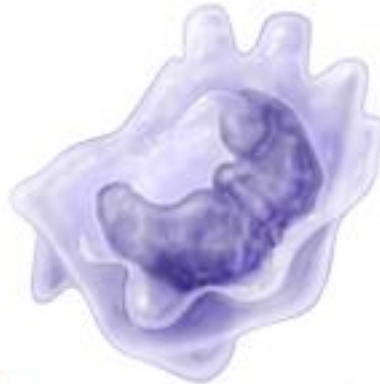
Banded neutrophil



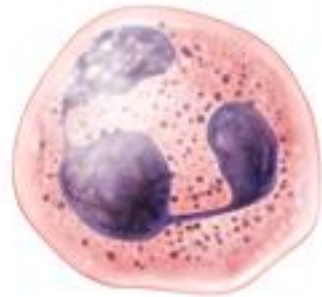
Eosinophil



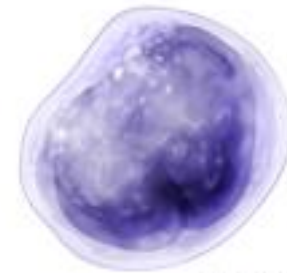
Monocyte



Neutrophil



Lymphocyte



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Leucocyte/WBCs

- Play an important role in the **immune system**
- **identifying, destroying, and removing pathogens, damaged cells, cancer cells, and foreign matter** from the body.
- produced within bones by bone marrow and some then mature in the **lymph nodes, spleen, or thymus gland**.
- The **only** blood components that are complete cells.
 - contain **nucleus**, and other **organelles**
- Less numerous than RBCs and platelets.

Leukocytes (WBCs)...

- WBC count :
 - Normal (4,000-11,000cell/mm³), Average (7,000 cell/mm³)
 - Decrease in WBC count = Leukopenia
 - A low WBC count may be due to disease, radiation exposure, or bone marrow deficiency
 - Increase in WBC count = Leukocytosis
- ✓ a normal, protective response to bacterial or viral invasion.
- A high WBC count may indicate the presence of an infectious or inflammatory disease, leukemia, stress or tissue damage.

Leukocytes (WBCs)...

- 1% of blood, but play vital role in the body's immune system.
- defense against invading **bacteria, viruses, fungi & parasites** by:

1. Direct attack [phagocytosis].

- identifying invading organism as foreign, attaching & destroying.

2. Produce antibodies into circulating blood to target & attach to foreign organisms.

- **neutralize** organism
- **elicit help** from other immune system cells to destroy foreign substance.

TABLE 14.4 | Abnormal White Blood Cell Numbers

White Blood Cell Population Change	Illness
Elevated lymphocytes	Hairy cell leukemia, whooping cough, mononucleosis
Elevated eosinophils	Tapeworm infestation, hookworm infestation, allergic reactions
Elevated monocytes	Typhoid fever, malaria, tuberculosis
Elevated neutrophils	Bacterial infections
Too few helper T cells (lymphocytes)	AIDS

Leukocytes (White Blood Cells)

Granular leukocytes

PMN-cells: Neutrophils

Eosinophils

Basophils

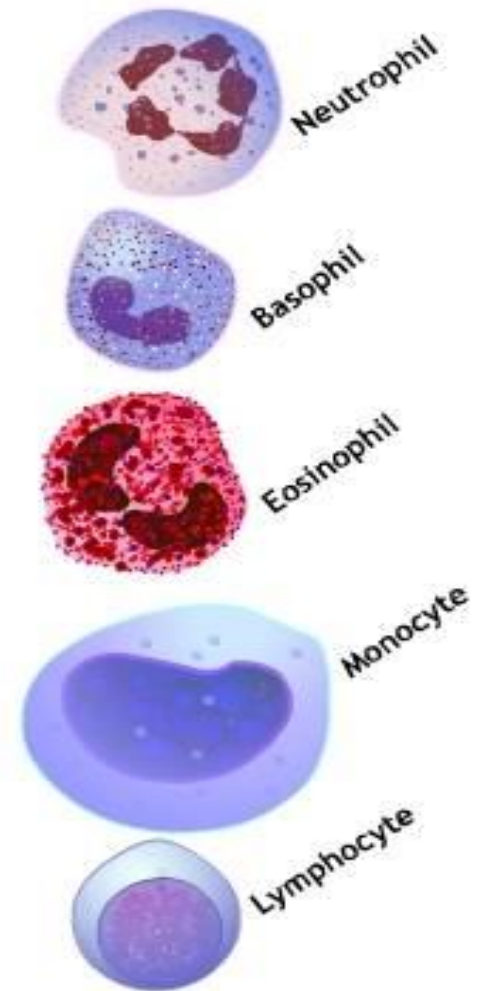
Agranular leukocytes

Lymphocytes

Monocytes

WBC differential count

- ❖ Neutrophils: 60-70% (62%), 3000-7000/mm³
- ❖ Eosinophils: 1-4% (2.4%), 100-440/mm³
- ❖ Basophils: 0.3-0.5% (0.4%), 20-50/mm³
- ❖ Monocytes: 2-8% (5.3%), 100-700/mm³
- ❖ Lymphocytes: 20-40% (30%), 1500-3500/mm³



Characteristics of WBCs

❑ **Mobility**: highly mobile & reach tissue fluids.

- **Diapedesis**

- squeezing out of WBC b/n capillary endothelial cells in to interstitial space.

- **Amoeboid motion**

- movement of WBC through tissue spaces by forming pseudopodia & reach microbes in tissues.

- **Chemotaxis**

- attraction of WBC towards injured or inflamed/infected area by chemicals or toxins produced by the microbe or inflamed tissues.

❑ **Phagocytosis**: engulfing & destroying microorganisms.

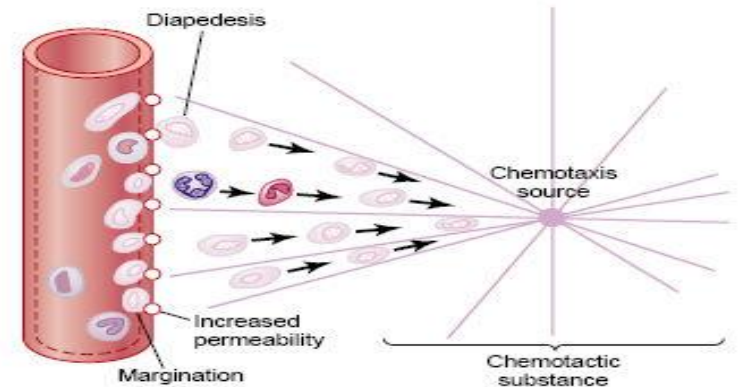
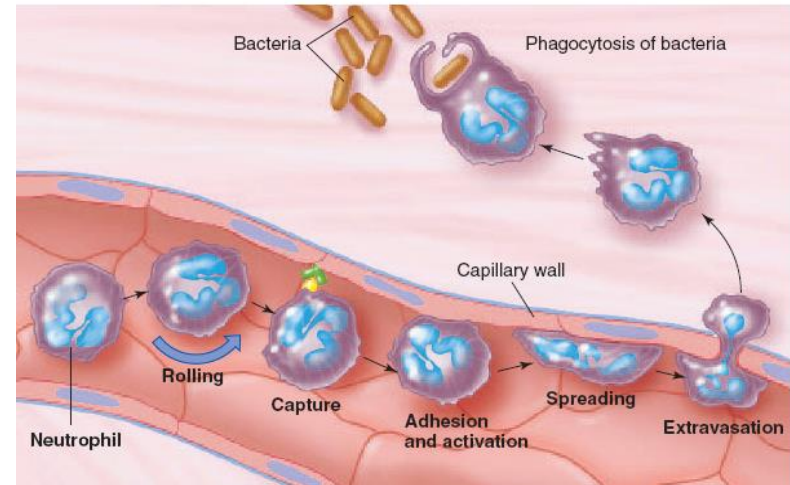


Figure 33-2

Movement of neutrophils by *diapedesis* through capillary pores and by *chemotaxis* toward an area of tissue damage.

Functions of WBCs

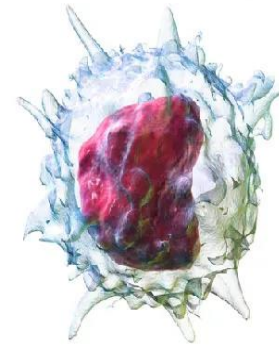
1. Neutrophils

- The most abundant (60-70%) & the first line of defense.
- Phagocytic cells (ingest bacteria or fungi), lysozyme.
- High number indicates bacterial infection.

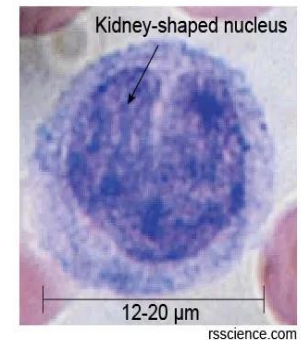


2. Monocytes

- The largest WBCs & highly phagocytic cells.
- High count may indicate viral/fungal infections, autoimmune disorder.
- ✓ Leave circulation, enter tissue & differentiate into macrophages:



Monocytes
(2 - 6% of WBCs)



The tissue macrophage system (reticuloendothelial system)

Monocytes are formed in the **bone marrow**



enter the **circulation**



leave the circulation & enter the **tissue**



↑**size**, ↑**lysosomal activities**



become tissue **macrophages**

Lungs

Skin

Liver

Brain

Bone

Spleen

**Alveolar
macrophage**

**Histocytic
cells**

**Kupffer
cells**

**Microglial
cells**

Osteoclasts

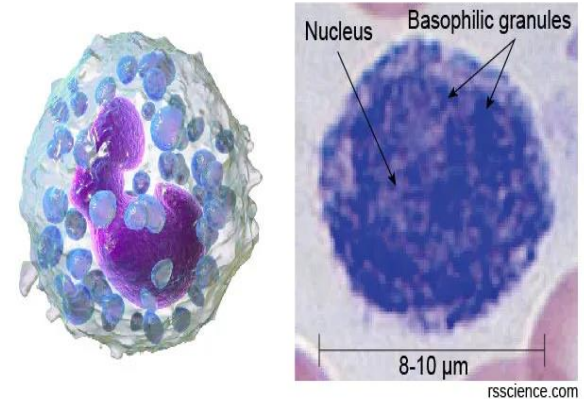
**Lymph nodes
reticular cells**

Functions of WBCs...

3. Basophils(least abundant)

- Liberate **heparin**, **histamine** & **serotonin** in allergic reactions that intensify overall inflammatory response.

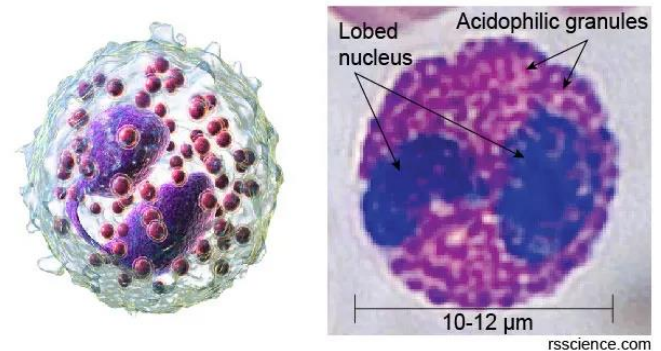
Basophils
(0.5 - 1% of WBCs)



4. Eosinophils

- Absorb **histamine** during allergic conditions so that **lessen severity of allergies**.
- Produce **hydrolytic enzymes** to kill big parasites. Increase in number during **parasitic infection**.
 - Eosinophilia

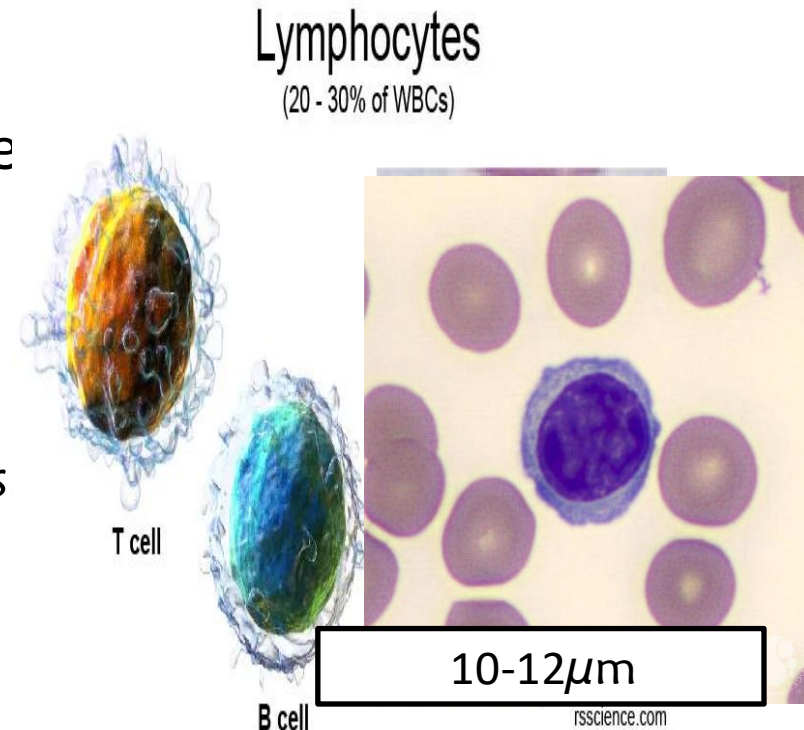
Eosinophils
(2 - 4% of WBCs)



Functions of WBCs...

5. Lymphocytes

- Are responsible for specific/
acquired immunity.
- Of 2 types; B and T lymphocyte
- An important cell class in the immune system that produces **antibodies:**
 - to **attack** infected & cancerous cells and
 - to **reject** foreign tissue.
- Their number increases in **viral infections.**



Formed Elements of the Blood

Component	Description	Number Present	Function
Erythrocyte (red blood cell)	Biconcave disc without nucleus; contains hemoglobin; survives 100 to 120 days	4,000,000 to 6,000,000 / mm ³	Transports oxygen and carbon dioxide
Leukocytes (white blood cells)		5,000 to 10,000 / mm ³	Aid in defense against infections by microorganisms
Granulocytes	About twice the size of red blood cells; cytoplasmic granules present; survive 12 hours to 3 days		
1. Neutrophil	Nucleus with 2 to 5 lobes; cytoplasmic granules stain slightly pink	54% to 62% of white cells present	Phagocytic
2. Eosinophil	Nucleus bilobed; cytoplasmic granules stain red in eosin stain	1% to 3% of white cells present	Helps to detoxify foreign substances; secretes enzymes that dissolve clots; fights parasitic infections
3. Basophil	Nucleus lobed; cytoplasmic granules stain blue in hematoxylin stain	Less than 1% of white cells present	Releases anticoagulant heparin
Agranulocytes	Cytoplasmic granules not visible; survive 100 to 300 days (some much longer)		
1. Monocyte	2 to 3 times larger than red blood cell; nuclear shape varies from round to lobed	3% to 9% of white cells present	Phagocytic
2. Lymphocyte	Only slightly larger than red blood cell; nucleus nearly fits cell	25% to 33% of white cells present	Provides specific immune response (including antibodies)
Platelet (thrombocyte)	Cytoplasmic fragment; survives 5 to 9 days	130,000 to 400,000 / mm ³	Enables clotting; releases serotonin, which causes vasoconstriction

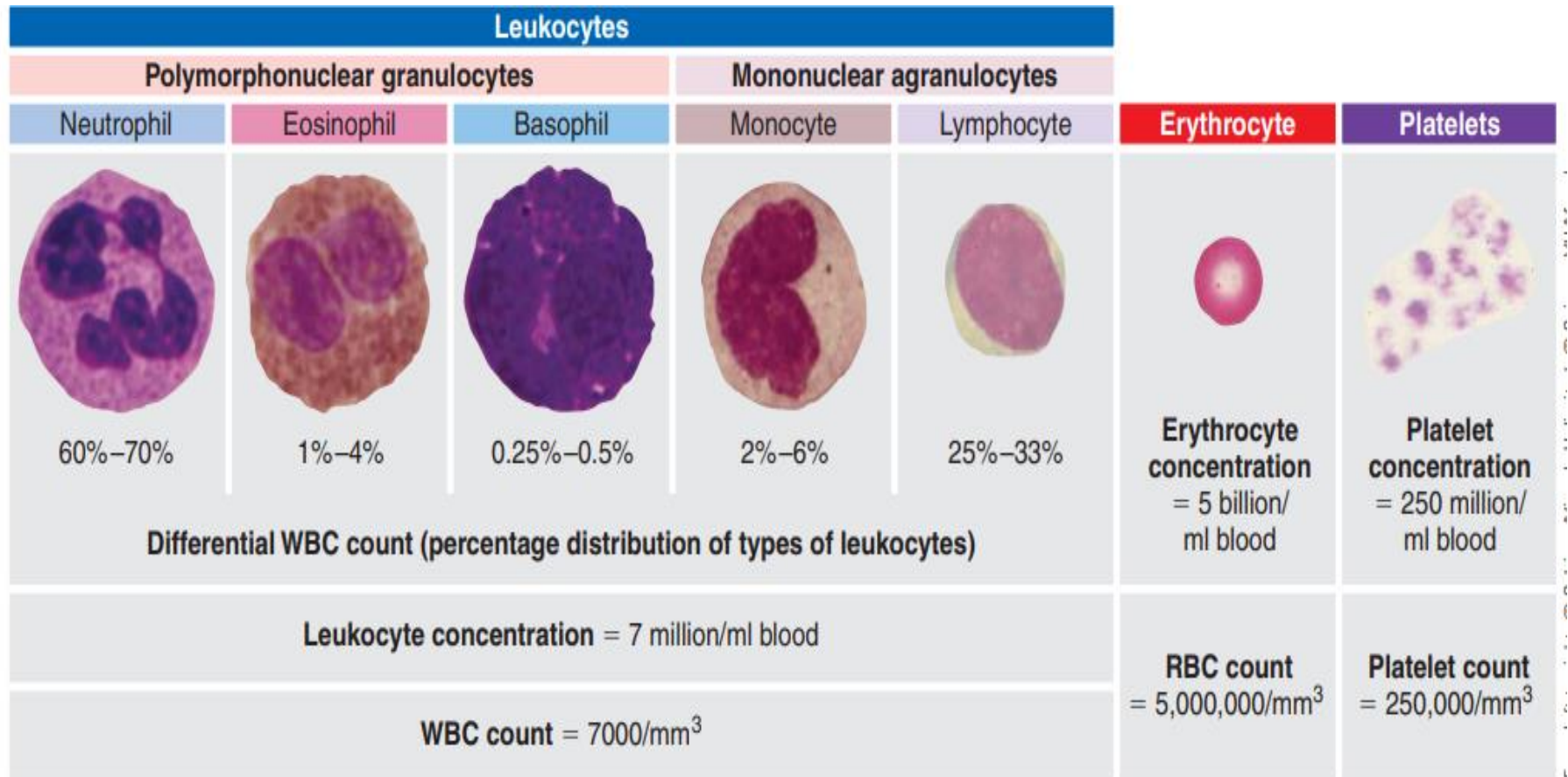


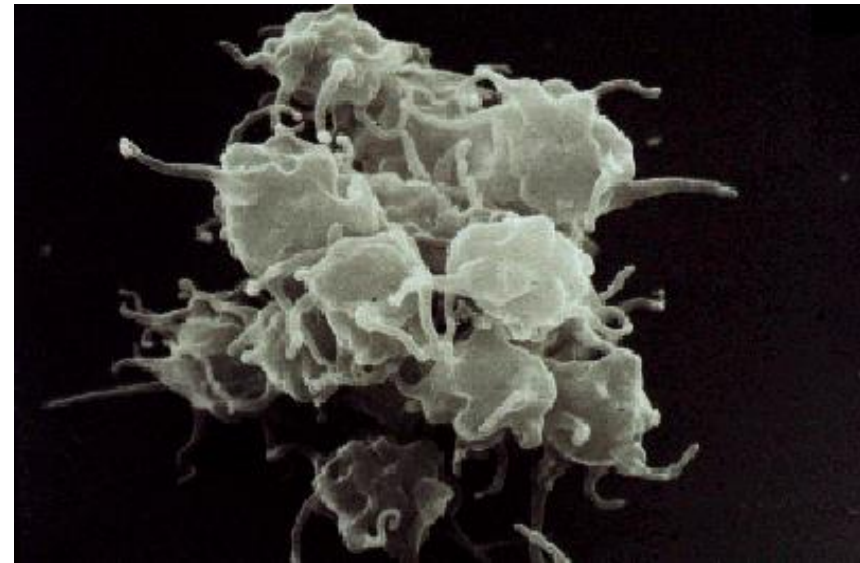
FIGURE :Normal blood cellular elements and typical human blood cell count.

Platelets

- Smallest fragment of cells in blood designed for coagulation
 - *forming clot* whenever blood vessel is broken.
- As soon as artery/vein is injured,
 - *platelets in area of injury begin to clump together & stick to edges of the cut.*



Inactive platelets



Active platelets

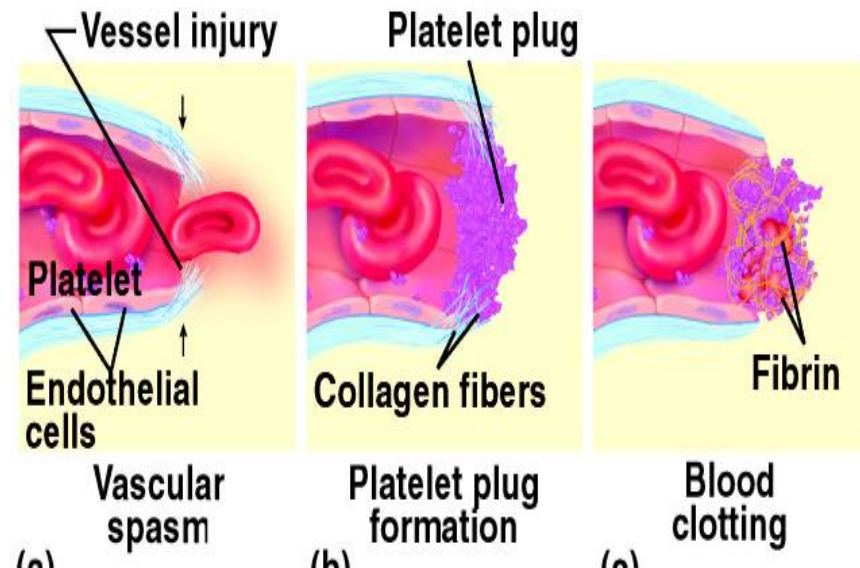
General Characteristics

- Have granules which releases chemicals up on stimulation;
 - serotonin, thromboxane, ADP, Ca^{++} , PDGF
- Contractile proteins *actin* & *myosin* help platelets to contract.
- Residuals of ; *ER, Golgi apparatus , Mitochondria*
 - synthesize enzymes, ATP & ADP
- Cell membrane has surface coat of *glycoproteins*:
 - repulses adherence to normal endothelium
 - adherence to *injured* endothelial cells (exposed *collagen*).
- Eliminated from circulation by tissue macrophage system.

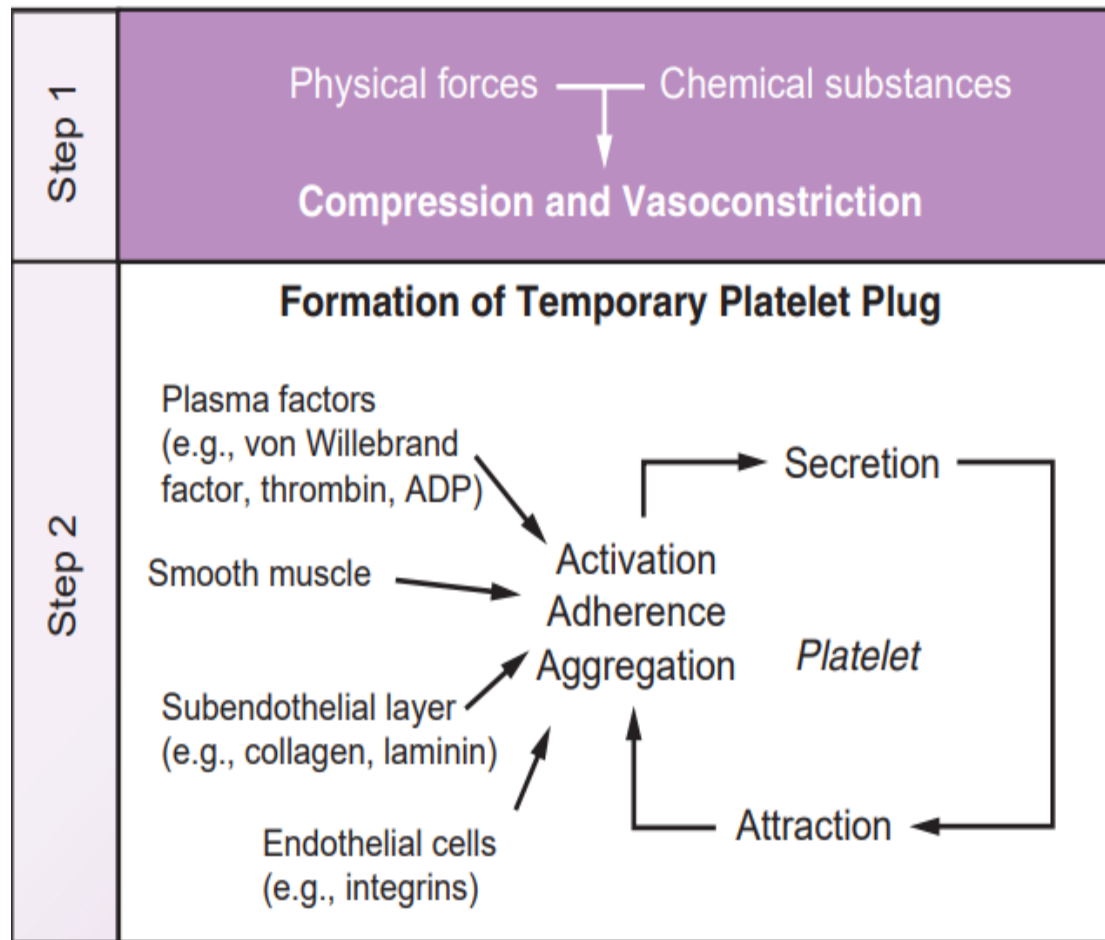
Hemostasis

- Process of **forming clots** in the walls of damaged blood vessels and preventing blood loss.
- Involves a series of reactions of plasma **coagulation factors** & **substances** released by platelets & injured tissues
- ✓ Can be organized into four separate but interrelated events:

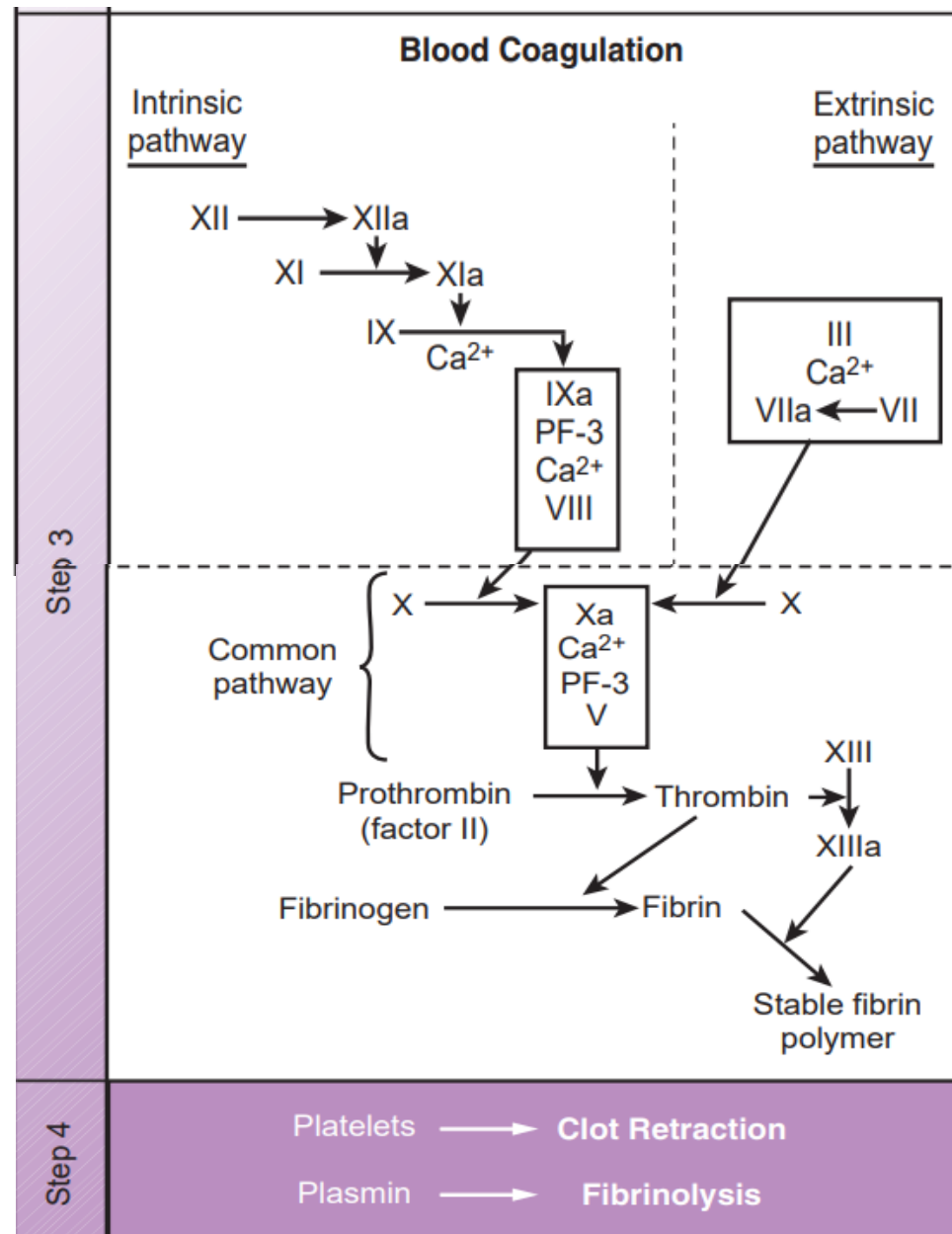
1. Compression and vasoconstriction;
2. Formation of a temporary loose platelet plug (primary hemostasis);
3. Formation of the more stable fibrin clot (secondary hemostasis),
4. Clot retraction and dissolution



Hemostasis. The blood's response to blood vessel injury can be viewed as four interrelated steps.

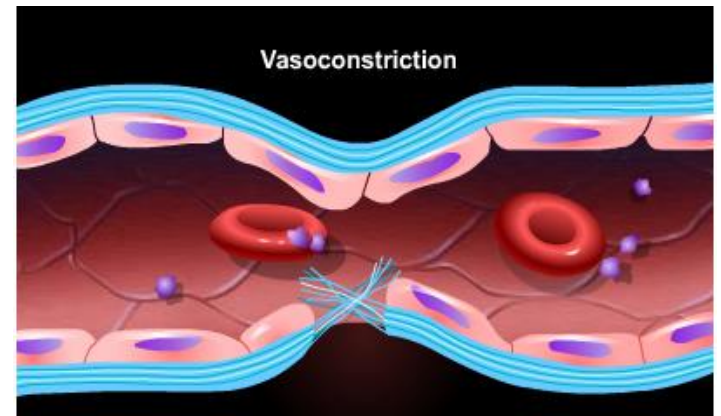
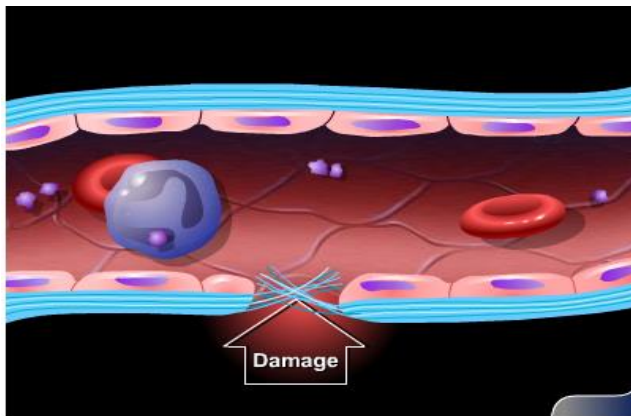


Cont'd....



Vascular spasm

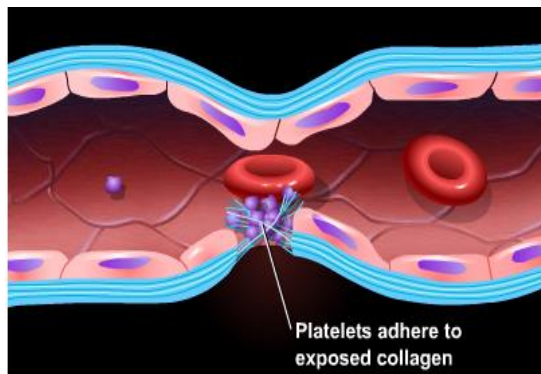
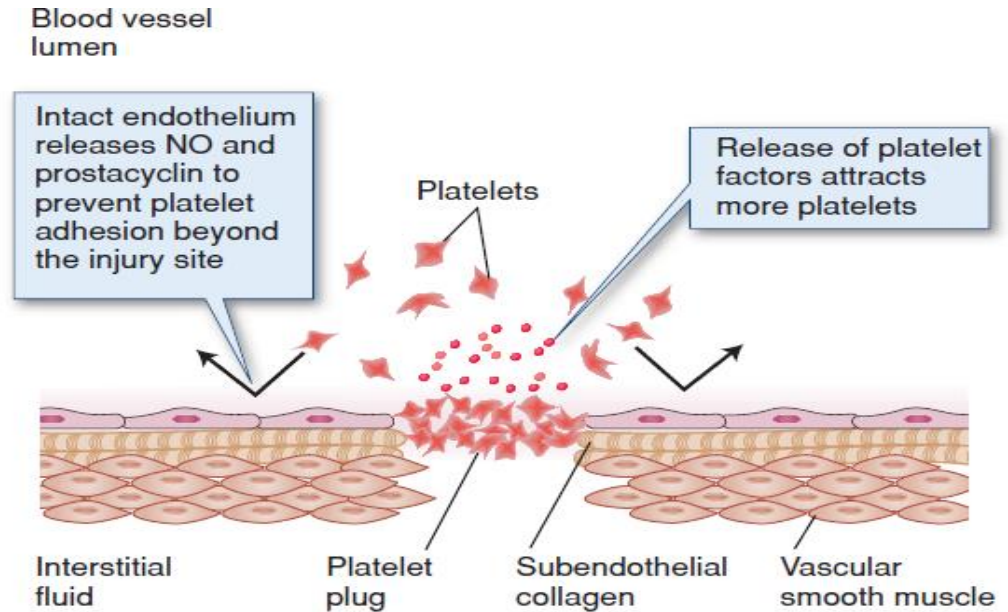
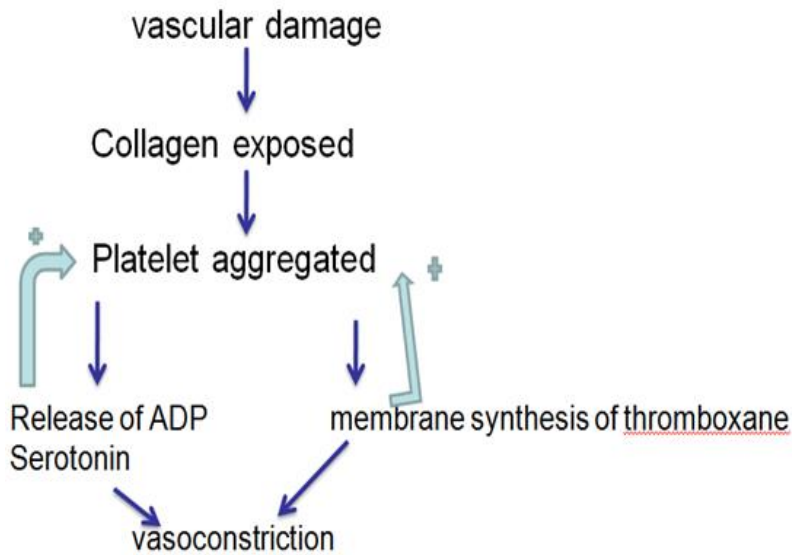
- Occurs immediately after blood vessel has ruptured.
- Local **vasoconstriction** that results from:
 1. Local **myogenic contraction** of vascular wall.
 - trauma to vessel wall causes smooth muscle in the wall to contract, instantaneously **reduces flow of blood** from the ruptured vessel.
 2. **Vasoconstrictor chemicals** released by damaged tissues (endothelin) & platelets([thromboxane A2, serotonin]).
 3. **Neuronal reflex** initiated by pain nerve impulses.



Platelet Plug Formation

- **1^{ry} hemostasis**, temporarily seals the break in the vessel wall.
- Injury to a vessel disrupts the endothelium and exposes the underlying connective-tissue collagen fibers.
- **platelets** adhere to collagen fibers with the help of adherence proteins (integrins) and large plasma protein (Von Willebrand Factor (VWF))
- VWF enhances **platelet adhesion** to the endothelium by forming a **bridge** b/n platelet surface receptors and collagen.
- When **platelet binds injured endothelial cells & collagen** they will undergo changes:
 - begins to swell, **become sticky**
 - contract forcefully to release multiple **active factors (ADP, serotonin & thromboxane A₂)**
 - Serotonin - vasoconstrictor
 - ADP - attracts more platelets to the area
 - Txa A₂ - promotes platelet aggregation & vasoconstriction

Mechanism of Platelet aggregation , positive feedback



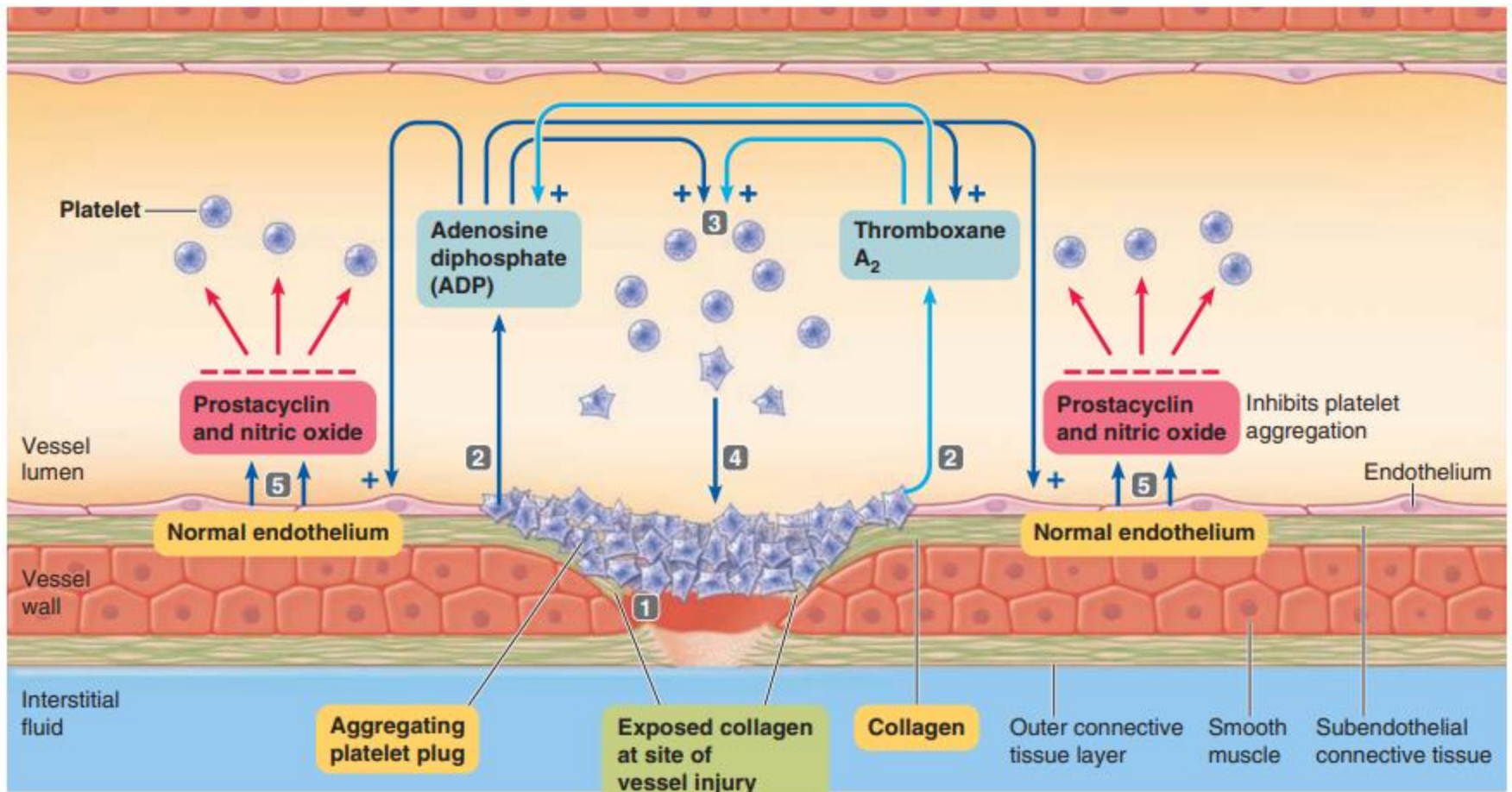
adhesion



activation



aggregation



1 Platelets adhere to and are activated by exposed collagen at the site of vessel injury.

2 Activated platelets release ADP and thromboxane A₂.

3 These chemical messengers work together to activate other platelets passing by.

4 Newly activated platelets aggregate onto growing platelet plug and release even more platelet-attracting chemicals.

5 Normal (uninjured) endothelium releases prostacyclin and nitric oxide, which inhibit platelet aggregation, so platelet plug is confined to site of injury.

Blood coagulation

- **2^{ry} hemostasis**, **slow** but **long lasting effect** to stop bleeding
- is the transformation of blood from a liquid form into a solid gel by fibrin.
- Formation of a clot on top of the platelet plug **strengthens and supports the plug**, reinforcing the seal over a break in a vessel.
- The fundamental reaction is conversion of **the soluble plasma protein fibrinogen to insoluble fibrin** by thrombin.
- Regulated by the **sequential activation of coagulation factors** in events called the **coagulation cascade**.

Mechanism of blood coagulation

- More than 50 important substances that **cause or affect blood coagulation** have been found in the blood and in the tissues.
- Factor that promote coagulation called **procoagulants**, and factor that inhibit coagulation, called **anticoagulants**.
- Coagulation depends on balance between these two groups of substances.
- **Anticoagulants predominate** in **intact vessel**, so the blood does not coagulate while its circulate in the blood vessels.
- When vessel is **ruptured**, **procoagulant activity increases** dramatically & clot formation begins.
-

Clotting factor/procoagulant

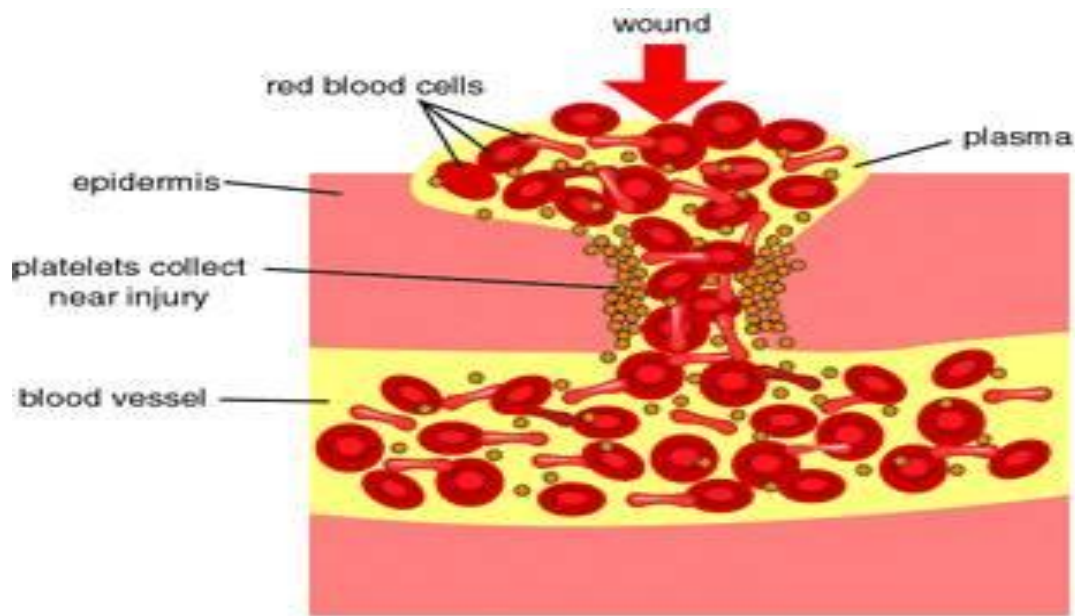
- Numbered I - XIII according to order of their discovery.
 - hence the numerical order does not reflect the reaction sequence.
- Most of these factors are **plasma proteins synthesized in the liver** and circulate in inactive form in blood until mobilized.
- **Vitamin K** is not directly involved in coagulation, but required for synthesis of 4 **procoagulants** made by **liver**. (Factor II, VII, IX, X)

Factors of the Coagulation Cascade

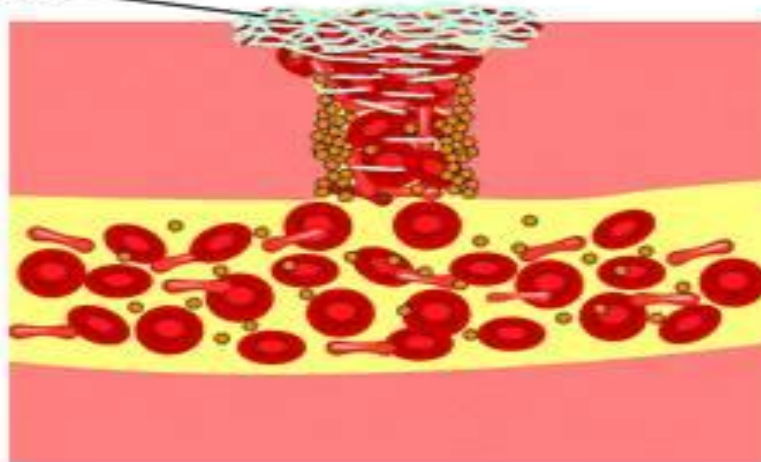
Scientific Name	Common Name	Other Names
Factor I	Fibrinogen	
Factor II	Prothrombin	
Factor III	Tissue thromboplastin	Tissue factor
Factor IV	Calcium	
Factor V	Proaccelerin	Labile factor
Factor VII	Proconvertin	Serum prothrombin conversion accelerator (SPCA)
Factor VIII	Antihemophilic factor	Platelet cofactor 1
Factor IX	Christmas factor	Platelet thromboplastin component
Factor X	Stuart factor	
Factor XI	Plasma thromboplastin antecedent	
Factor XII	Hageman factor	Contact factor
Factor XIII	Fibrin stabilizing factor	

Blood clotting/coagulation takes place in three essential steps(general mechanism):

1. In response to rupture of the vessel or damage to the blood itself, a complex cascade of chemical reactions occurs resulting **formation** of **prothrombin activator** through **intrinsic & extrinsic pathways**.
2. **Conversion** of **prothrombin into thrombin** by action of prothrombin activator.
3. **Conversion** of **fibrinogen into fibrin thread** by action of thrombin



red blood cells and platelets trapped in fibrin mesh form clot



injured tissues and platelets release the clotting factor **prothrombin activator** and **calcium ions**



prothrombin activator converts the blood protein prothrombin to **thrombin**



thrombin splits **fibrinogen** to form **fibrin**



fibrin fibres form a mesh over wound, trapping red blood cells and platelets



bleeding stops



clot hardens and becomes smaller



new cells grow to repair wound site



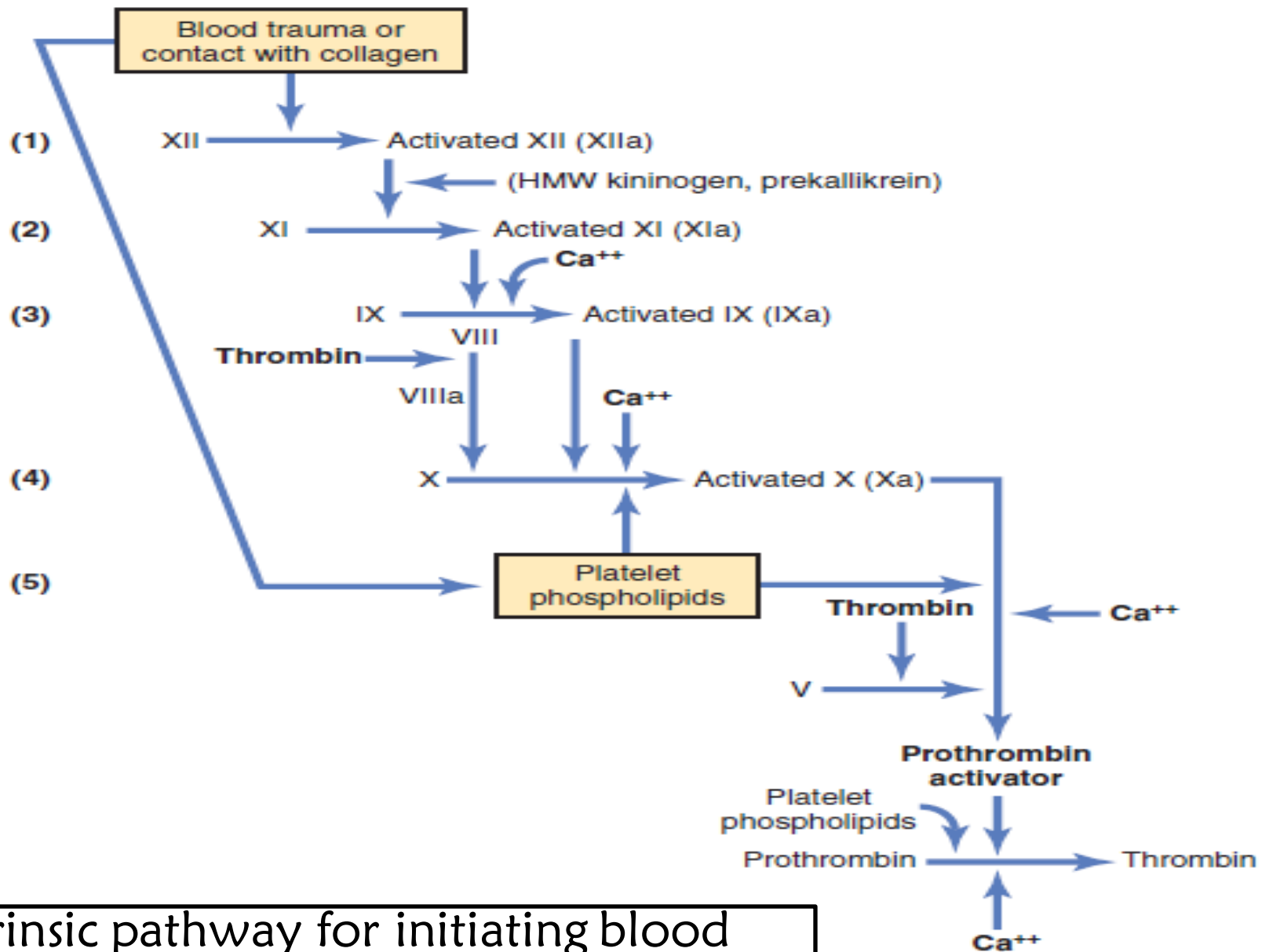
enzyme **plasmin** is released to dissolve clot

Formation of prothrombin activator (Initiation of Coagulation)

- Clotting can be initiated by either **intrinsic/extrinsic pathway**.
- In the body, both pathways are usually triggered by **injury to the blood & blood vessel**.
 - **Chemicals** released by injured vascular wall & adjacent tissues initiates **extrinsic pathway**.
 - **Trauma** to blood or blood contact with damaged endothelium/collagen initiates **intrinsic pathway**
- Clotting of blood outside the body (test tube) is initiated only by **intrinsic mechanism**.

Intrinsic pathway

- Slower.
- Triggered by blood trauma or when **blood contacts foreign surface**, injured endothelial cells/collagen or test tube, causing activation of **Factor XII and release of platelet phospholipids**
- Factor **XII** activates factor **XI** which in turn activates factor **IX**
 - ✓ **require** high-molecular-weight kininogen and is accelerated by prekallikrein).
- Activated **factor IX** joins with factor **VIII**, **Ca⁺⁺**, platelet phospholipids to activate **factor X**.

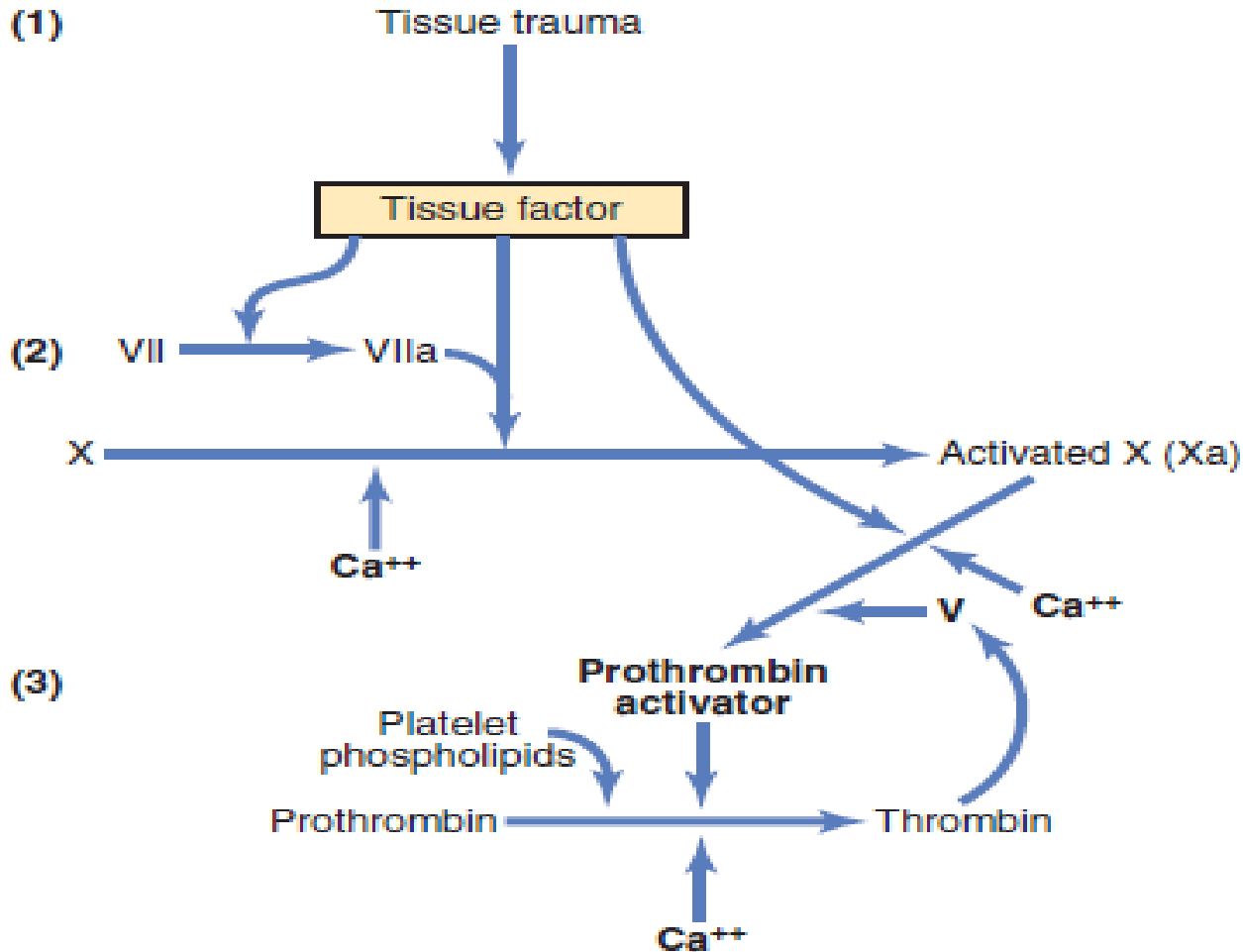


Intrinsic pathway for initiating blood clotting.

Extrinsic pathway

- Faster.
- Triggered when traumatized tissue releases a **tissue factor or tissue thromboplastin (factor III)**.
- The lipoprotein complex of tissue factor further complexes with Factor VIIa to activate Factor X (Xa) .
- ✓ requires ionic **calcium**
- The activated Factor X combines with Factor V, to form the complex called **prothrombin activator** in the presence of Ca^{++} and tissue/platelet phospholipids

Extrinsic pathway for initiating blood clotting.

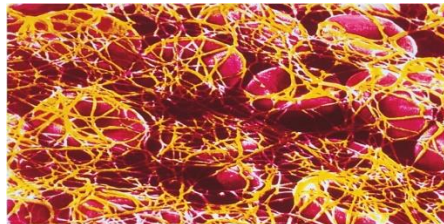


Common pathway

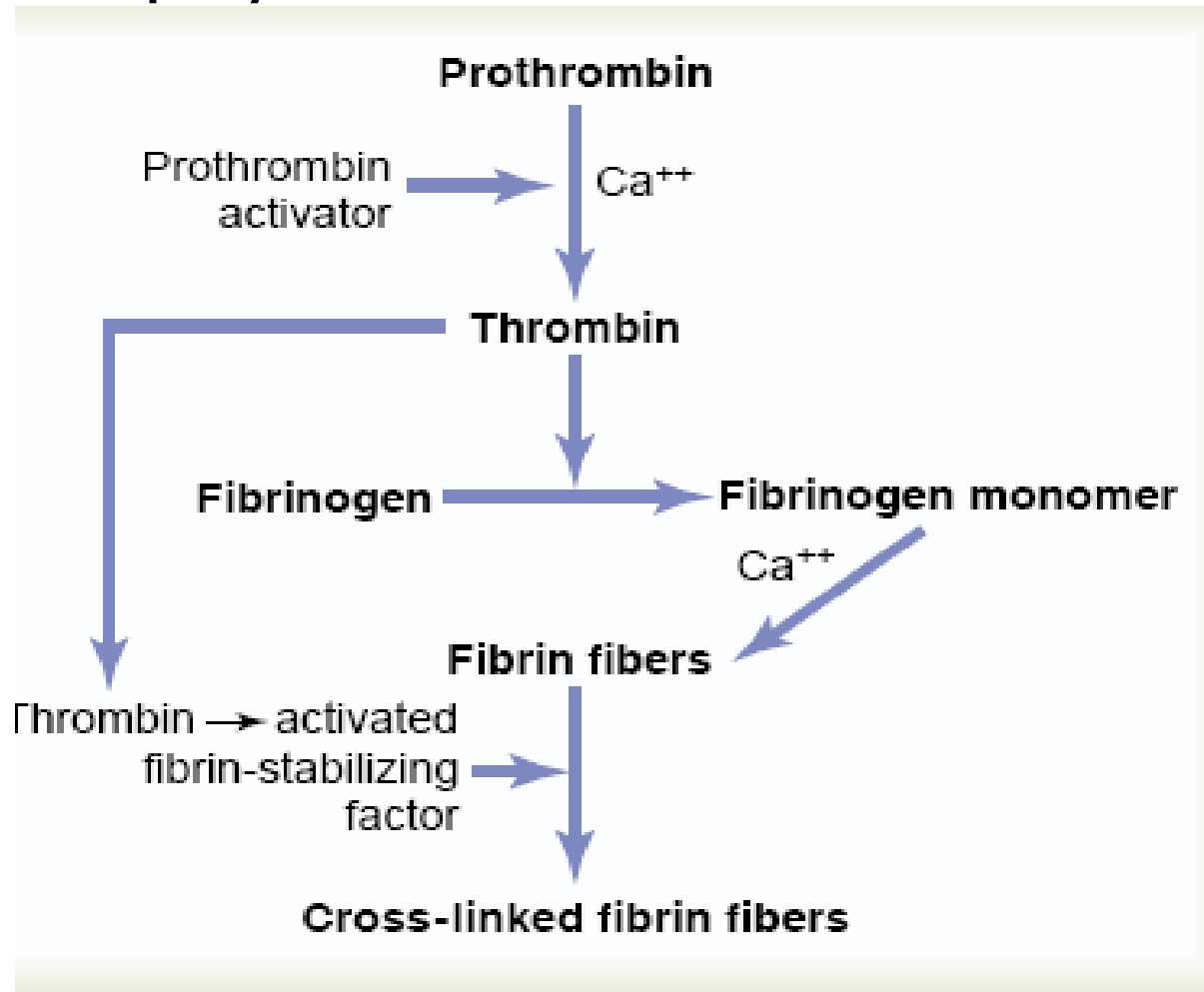
- The final events leading to **fibrin formation** by both **intrinsic and extrinsic pathways**
- It is initiated by the conversion of inactive clotting **factor X** to its active form, **factor Xa**
- Activated factor X complexes with **Ca⁺⁺**, platelet phospholipids & **factor V** to form **prothrombin activator**.
 - Process of forming prothrombin activator is the slowest step of blood clotting process.
- Prothrombin activator converts **prothrombin** to its active form **thrombin**.

Common pathway ...

- **Thrombin** is a protein enzyme that remove 4 peptides from fibrinogen forming insoluble fibrin monomer.
- Fibrin monomers spontaneously assemble into ordered fibrous arrays of fibrin strands forming a highly organized and firm network(web),
- ✓ which traps **more platelets, red blood cells, and leukocytes** at the site of vascular damage, thereby forming a **stable blood clot**.
- A plasma enzyme, **fibrin stabilizing factor (factor XIII)**, catalyzes the formation of **covalent bonds b/n strands of polymerized fibrin**, stabilizing and tightening the blood clot



Conversion of prothrombin to thrombin & polymerization of fibrin fibers



Extrinsic pathway

Activator:
tissue thromboplastin

VII → VII activated

VII complex
(VII, tissue
thromboplastin,
calcium,
phospholipids)

Intrinsic pathway

Activators:
collagen, glass,
and others

XII → XII activated

XI → XI activated

IX → IX activated

VIII complex
(VIII, IX activated,
calcium, phospholipids)

Common pathway

X → X activated

V complex
(V, X activated,
calcium,
phospholipids)

Prothrombin → Thrombin

Fibrinogen → Fibrin

Blood Physiology

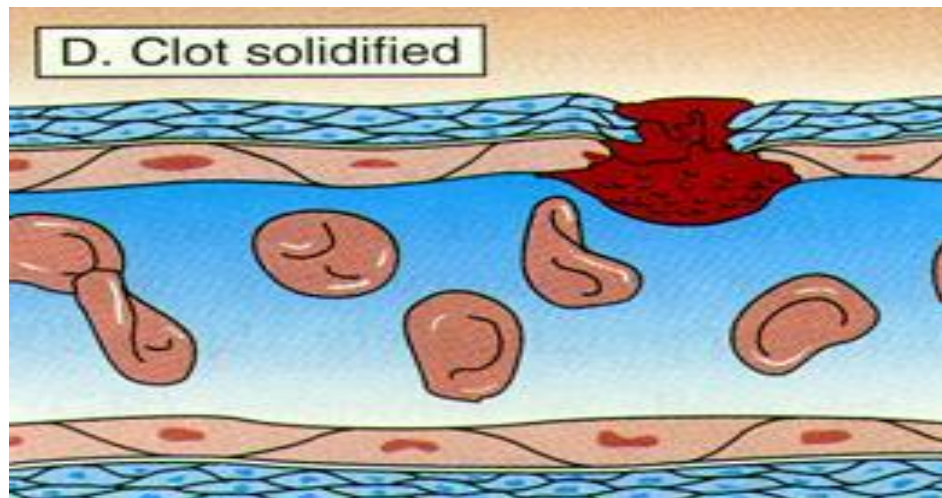
XIII

Fibrin
polymer

Clot retraction & repair

□ Clot retraction

- Further stabilization of clot by squeezing serum from fibrin strands.
- after clot is formed, it begins to contract (*platelets*) & usually expresses most of fluid (*serum*) from the clot.
- as clot retracts, ruptured edges of the blood vessel slowly brought together again to **repair** the damage.



□ Vessel Repair

- Once blood clot has formed it is invaded by **fibroblasts** (form a **tissue patch**).
- This is promoted partially by growth factor secreted by platelets
- **Platelet derived growth factors(PDGF)** stimulates smooth muscle cells, vascular endothelial cells & fibroblasts to divide & rebuild the wall.

Clot dissolution

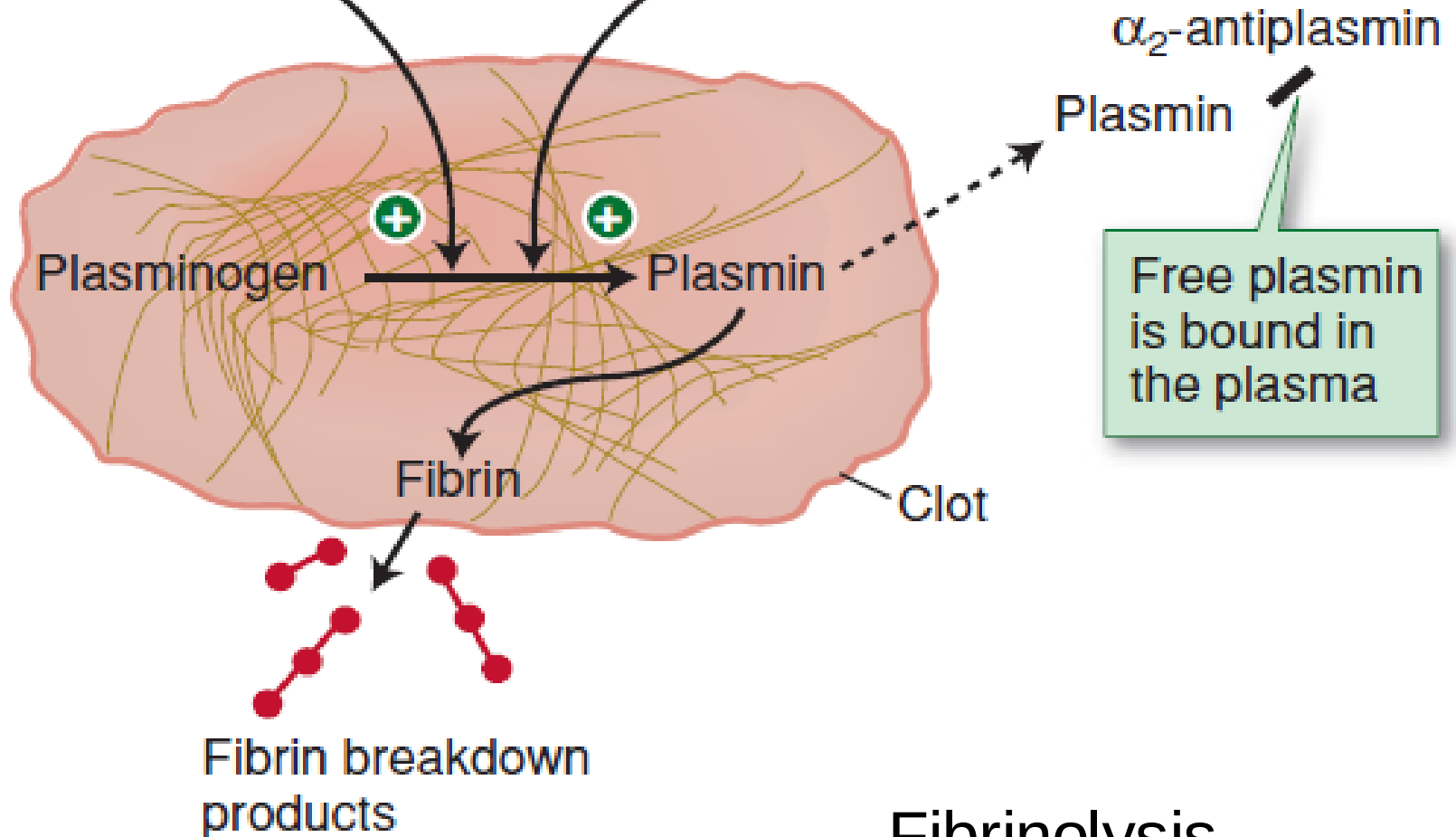
- Once the wall of damaged blood vessel is repaired, the blood clot is removed by **lysis**.
- **Fibrinolysis** removes unneeded clots when healing has occurred
- without fibrinolysis, blood vessels would gradually become completely blocked.
- fibrinolysis begins within 2 days & continues slowly over several days until the clot is finally dissolved.

Plasmin cause clot dissolution

- **Plasmin (fibrinolysin)** “fibrin splitting” enzyme, dissolve the clot.
- Plasmin is a plasma protein produced by the liver and present in the blood in an inactive precursor form, plasminogen.
- Plasminogen activators include:
 - **Factor XII** (Hageman Factor), **tissue plasminogen activator** (tPA) from vascular endothelial cells & **thrombin**.
- Plasmin gets trapped in clot & slowly dissolves it by breaking down the fibrin meshwork.
- Phagocytic white blood cells gradually remove the products of clot dissolution.

Tissue plasminogen activator

Urokinase-type plasminogen activator



Fibrinolysis

Prevention of unwanted intravascular clotting

1. **Smoothness** of endothelial cell surface
 - prevents contact activation of intrinsic clotting system
2. Layer of **glycocalyx** on vascular endothelium
 - repels platelets & clotting factors
3. Presence of **thrombomodulin** which binds thrombin
 - removes thrombin
4. **Heparin**
 - inactivate thrombin, IXa, Xa, XIa
 - activate other anticoagulants (anti-thrombin III)
5. **Antithrombin III**, inactivates thrombin and several other clotting factors

Factors That Inhibit Blood Clot Formation

Factor	Action	Factor	Action
Smooth lining of blood vessel	Prevents activation of intrinsic blood clotting mechanism	Antithrombin in plasma	Interferes with the action of thrombin
Prostacyclin	Inhibits adherence of platelets to blood vessel wall	Heparin from mast cells and basophils	Interferes with the formation of prothrombin activator
Fibrin threads	Adsorbs thrombin		

6. Nitric oxide & prostacyclin

– released by intact vascular endothelial cells, prevents platelet aggregation.

7. Plasmin /fibrinolysin

- break down fibrin into fibrinogen, lysis the clotted ones
- digest fibrin & other clotting factors (fibrinogen, prothrombin, V,VIII,XII)
- important in the blood to avoid occlusion

Abnormal intravascular clotting

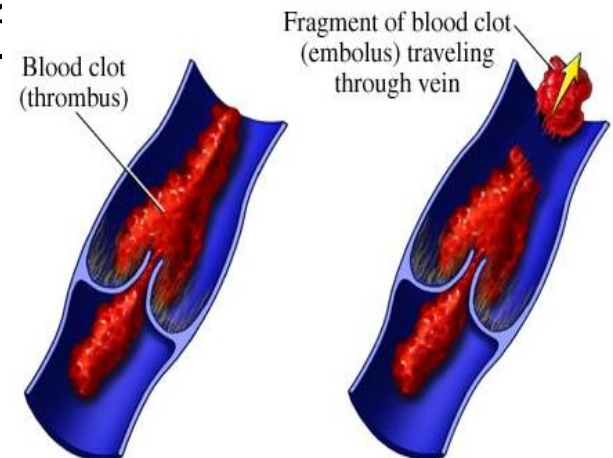
Thromboembolic Conditions

Thrombus – clot that develops & persists in uninjured blood vessel

- Thrombi can **block flow in the circulation**, resulting in tissue death & organ failure.
- **Coronary thrombosis** – thrombus in blood vessel of heart
e.g. Myocardial infarction (heart attack)

Embolus - thrombus (blood clot) freely f

- **Pulmonary emboli** can impair the ability of the body to obtain oxygen.
- **Cerebral emboli** can cause strokes.



- Cause of Thromboembolic Conditions ;
 - A roughened endothelial surface of a vessel—as may be caused by arteriosclerosis, infection, or trauma
 - Sluggishness of blood flow.
 - Increase in viscosity of blood due to high platelet or high RBC count.
 - Widespread clotting is occasionally triggered by the release of tissue thromboplastin into the blood from large amounts of traumatized tissue.
- Similar widespread clotting can occur in septicemia, in which bacteria or their toxins(endotoxin) initiate the clotting cascade

Erythrocyte Disorders

I. Anemia

- Insufficient Erythrocytes, decreased hemoglobin content.
 - blood has abnormally low O_2 carrying capacity.
- Blood O_2 levels can not support normal metabolism.

II. Polycythemia

- Excess RBCs production, of 2 types:
 1. Primary polycythemia (polycythemia Vera)
 - bone marrow abnormality.
 2. Secondary polycythemia
 - mostly physiologic, 2ry to hypoxia (higher altitude, EPO)

Leukocytes Disorders

I. Leukemia

- Abnormal increase of white blood cells.
- Cancerous conditions causing proliferation of immature WBCs

II. Leukopenia (leukocytopenia)

- Decrease number of WBCs (leukocytes), which places individuals at increased risk of infection.

Bleeding disorders

- **Inability** to synthesize **procoagulants** (clotting factors) by **liver** results in severe bleeding disorders.
- Bleeding can be caused by:
 - Hepatitis/cirrhosis
 - Vitamin K deficiency
 - **Hemophilia**
 - Thrombocytopenia (platelet deficiency)
 - **purpura**

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